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Serial ATA Interoperability Program Revision 1.5 Teledyne LeCroy Method of Implementation (MOI) Document for PHY, TSG & OOB Tests (Real-time DSO measurements)

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MODIFICATION RECORD

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TEMPLATE INITIAL RELEASE, TO LOGO TF MOI GROUP

February 22, 2006 v0.9
LeCroy SDA11000 MOI, including updates from IW#1

September 14, 2006
change to update tests that have changed, and to use QualiPHY test suite.

September 27, 2006
edit TSG-01 to include pu/pl as informative.

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add oscilloscope specifications table

November 17, 2006
add more description of test implementation

January 9, 2007
TSG-06 change to statistical mode

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Change OOB-01, OOB-06 and OOB-07 to use AWG
Add lab load return loss verification procedure.
Change to Comax fixture

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Updated Logo

November 5, 2007
Changes in PHY-02 and PHY-04 for ECN 16
Changes in OOB-02 – OOB-05 for ECN17
Add Appendix E with JMD calibration procedure
Update references for Rev. 2.6 of the SATA spec

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Add note to use JTF settings for TSG-09 – TSG-12

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Improve instructions for hosts for OOB tests
Add instrument specific instructions for JMD Calibration
Replace Gen1 and Gen2 references with bitrates

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Add Appendix F with amplitude calibration procedure for OOB-01
Add LeCroy document number
Remove reference to ECN017 from OOB-02
Remove group numbers for PHY, TSG, OOB sections

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July 17, 2008 (Rev 1.3 version 1.0)
Removed RC

March 25, 2009 (Rev 1.4 version 0.8)
Update references for SATA 3.0 spec and UTD 1.4
Update resource requirements with new models
Modify TSG-02 and TSG-5 to use LFTP and add 6.0Gb/s limits
Remove obsolete tests TSG-07 and TSG-08
Add TSG-13 – TSG-16
Update JTF procedure for Gen3

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Reviewed by SATA Logo group and changed version

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Changed TSG-05 and TSG-06 to Obsolete
End 30 day review –release as 1.0

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Updated required firmware versions
Add ECN39 to TSG-13
Add LeCroy TF-SATA-C and Wilder fixtures in Appendix A
Add LeCroy PeRT^3 and Sierra as a BIST Init Tools in Appendix A
Add instructions for PeRT^3 BIST-L in Appendix C

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Reviewed by SATA Logo group and changed version

April 29, 2011 (Rev 1.4.2 Version 0.9)
Updated required firmware versions
Modified TSG-15 to use 5MUI eye height instead of 1e-12 BER
Modified Appendix C for new Host OOB timing and OOB testing with the PeRT^3
Modified OOB tests for use with PeRT^3
Added PeRT^3 and Sierra M6 as BIST initiator tools in test resources

May 5, 2011 (Rev 1.4.2 Version 0.91)
Modified OOB-06 timing

May 12, 2011 (Rev 1.4.2 Version 1.0RC)
Approved by SATA Logo group as release candidate

May 6, 2013 (Rev 1.5 Version 0.8)
Updated references from LeCroy to Teledyne LeCroy
Modified oscilloscope channels being used
Modified TSG-02 and TSG-03 to be informative
Updated TSG-02 rise/fall time limits
Modified TSG-04 required test patterns and limits
Modified TSG-13 to remove the CIC when testing UHOST (Gen 3u) and removed references to Rj
Modified TSG-15 to remove the CIC when testing UHOST (Gen 3u)
Modified TSG-16 to be obsolete
Updated oscilloscope models in Appendix A

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Added TSG-17 Transmitter Emphasis section
Adjusted references to reflect the 1.5 UTD

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INTRODUCTION

The tests contained in this document are organized in order to simplify the identification of information related to a test, and to facilitate in the actual testing process. Tests are separated into groups, primarily in order to reduce setup time in the lab environment, however the different groups typically also tend to focus on specific aspects of product functionality.

The test definitions themselves are intended to provide a high-level description of the motivation, resources, procedures, and methodologies specific to each test. Formally, each test description contains the following sections:

Purpose

The purpose is a brief statement outlining what the test attempts to achieve. The test is written at the functional level.

References

This section specifies all reference material *external* to the test suite, including the specific subclauses references for the test in question, and any other references that might be helpful in understanding the test methodology and/or test results. External sources are always referenced by a bracketed number (e.g., 1) when mentioned in the test description. Any other references in the test description that are not indicated in this manner refer to elements within the test suite document itself (e.g., “Appendix 6.A”, or “Table 6.1.1-1”)

Resource Requirements

The requirements section specifies the test hardware and/or software needed to perform the test. This is generally expressed in terms of minimum requirements, however in some cases specific equipment manufacturer/model information may be provided.

Last Modification

This specifies the date of the last modification to this test.

Discussion

The discussion covers the assumptions made in the design or implementation of the test, as well as known limitations. Other items specific to the test are covered here as well.

Test Setup

The setup section describes the initial configuration of the test environment. Small changes in the configuration should not be included here, and are generally covered in the test procedure section (next).

Procedure

The procedure section of the test description contains the systematic instructions for carrying out the test. It provides a cookbook approach to testing, and may be interspersed with observable results.

Observable Results

This section lists the specific observables that can be examined by the tester in order to verify that the Product Under Test (PUT) is operating properly. When multiple values for an observable are possible, this section provides a short discussion on how to interpret them. The determination of a pass or fail outcome for a particular test is generally based on the successful (or unsuccessful) detection of a specific observable.

Possible Problems

This section contains a description of known issues with the test procedure, which may affect test results in certain situations. It may also refer the reader to test suite appendices and/or other external sources that may provide more detail regarding these issues.

REFERENCES

The following documents are referenced in this text:

1. Serial ATA Revision standard, Revision 3.2
2. Serial ATA Interoperability Program Unified Test Document Revision 1.5
3. Serial ATA Interoperability Program Policy Document Revision 1.4
4. Serial ATA Interoperability Program Pre-Test MOI Revision 1.4

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PHY GENERAL REQUIREMENTS

Overview:

This group of tests verifies the Phy General Requirements, as defined in Section 3.14 of the SATA Interoperability Unified Test Document (which references the SATA Standard).

Test PHY-01 - Unit Interval

Purpose: To verify that the Unit Interval of the Product Under Test (PUT) TX signaling is within the conformance limits.

References:

[1] SATA Standard, 7.4.2, Table 52 – General Specifications
[2] Ibid, 7.4.3.1.4 – Unit Interval
[3] Ibid, 7.6.16 – SSC Profile
[4] SATA PRE-TEST MOI
[5] SATA Interoperability Program Unified Test Document, 3.14.1

Resource Requirements:

Teledyne LeCroy oscilloscope (see Appendix A for specific models) with 7.7.x.x or later firmware, with QPHY-SATA option key.
Teledyne LeCroy QualiPHY SATA test suite version 7.7.x.x.
SATA test fixture and 2 SMA cables.
BIST Initiator – One of the following or any other mechanism that makes the product produce the required patterns:
Teledyne LeCroy PeRT3 with SATA option, software version 2.70 or later.
Teledyne LeCroy Sierra M6-2 or M6-4 Protocol Test System with software version 3.40 or later.
Teledyne LeCroy SASTrainer Protocol Exerciser with SASTracer software version 2.80 or later.

See appendix A for details.

Last Modification: December 22, 2014

Discussion:

Reference [1] specifies the general PHY conformance limits for SATA products. This specification includes conformance limits for the mean Unit Interval (UI). Reference [2] provides the definition of this term for the purposes of SATA testing. Reference [3] defines the measurement requirements for this test. Reference [4] describes the procedure for placing the PUT into BIST mode to generate the required test patterns.

In this test, the mean UI is measured from an SSC Track. The SSC Track is created by applying an FM demodulator and a low pass filter with 1.98MHz -3dB bandwidth. A 500µs acquisition is used to generate the SSC Track containing at least 10 cycles. The slowest allowed SSC cycle is 30kHz, which is 33.333µs per cycle; so in 500µs there are at least 15.0 cycles of SSC. (At 32kHz f_{SSC} there are exactly 16.0 cycles of SSC modulation in 500µs. The fastest allowed SSC cycle is 33kHz, which is 30.303µs; at that f_{SSC} there are 16.5 cycles of SSC modulation in 500µs.).

This requirement must be tested at all supported interface rates (1.5Gb/s, 3.0Gb/s and 6.0Gb/s).

Test Setup:

Channels 1 and 2 of the oscilloscope are connected to the transmitter pair on the SATA test fixture. For devices this is 5 and 6, and for hosts it is 2 and 3. The SATA test fixture is inserted into the PUT. For hosts, this test should be done on the worst case port identified. See Reference [4] for details.

Test Procedure:

The channels should be deskewed before performing this test. See Appendix A for details. The test procedure then proceeds as follows:

1) Open the QualiPHY SATA PHY, TSG, OOB, RSG test suite. See Appendix B for details.

For products that support bitrates greater than 1.5Gb/s, steps 2-5 must be performed at all speeds:

2) Select an appropriate configuration for the speed being tested with PHY-01 selected and with the proper SSC setting.
3) Run the test.

4) If using a PeRT3, it will place the product in BIST and cause it to output the required patterns. Otherwise, when prompted to produce HFTP set the product to generate HFTP. See Appendix C for details. Check the pattern on the oscilloscope to make sure it is correct. Press "OK" to continue.
5) When the test completes, generate a report. Observe the results for PHY-01 in the report.

Observable Results:

- PHY-01a - Mean Unit Interval measured between 666.4333ps (min) to 670.2333ps (max) (for products running at 1.5Gb/s)
- PHY-01b - Mean Unit Interval measured between 333.2167ps (min) to 335.1167ps (max) (for products running at 3.0Gb/s)
- PHY-01c - Mean Unit Interval measured between 166.6083ps (min) to 167.5583ps (max) (for products running at 6.0Gb/s)

Possible Problems:

The following is applicable to all PHY and TSG tests:

If the product under test (PUT) supports BIST with T, A and S but drops out of BIST T when disconnected, then it will be necessary to use two SATA test fixture adapters (or equivalent) and four SMA cables to connect the PUT to the SASTracer/Trainer to put it into BIST T and then remove one of the SMA cables from the PUT transmitting side of the SATA test fixture at the SASTracer/Trainer and connect it to the oscilloscope; then do the same for the other SMA cable. The assumption is that the product will remain in BIST T if this procedure is followed.

If the product does not support BIST T but does support BIST L, it will be necessary to use either a signal generator or another product that does support BIST T as a source of the pattern; and place the product under test in BIST L (loopback) mode, and loop the pattern through it. If the product under test does not support disconnect without dropping out of BIST L, then it will be necessary to use two SATA test fixtures (or equivalent) and four SMA cables to connect the PUT to the SASTracer/Trainer to put it into BIST L and then remove one of the SMA cables from the PUT transmitting side of the SATA test fixture at the SAS Tracer/Trainer and connect it to the oscilloscope; then do the same for the other SMA cable; and also remove first SMA cable from the PUT receiving side of the SATA test fixture connected to the SASTracer/Trainer and connect it to the matching SMA connector on the SATA test fixture connected to the device sourcing the pattern; then do the same for the last SMA cable connected to the SAS Tracer/Trainer.

The user is encouraged to always make sure that the correct pattern is being produced. The SATA power cable can sometimes be sensitive to small shifts causing the power to be reset so that the product will no longer be transmitting the power. Therefore, the user should be careful to avoid this when switching connections.

Test PHY-02 – Frequency Long Term Accuracy

Purpose: To verify that the long term frequency accuracy of the PUT's transmitter is within the conformance limit.

References:

[1] SATA Standard, 7.4.2, Table 52 – General Specifications
[2] Ibid, 7.4.3.1.5 – TX Frequency Long Term Accuracy
[3] Ibid, 7.6.9 – Long Term Frequency Accuracy
[4] SATA PRE-TEST MOI
[5] SATA Interoperability Program Unified Test Document, 3.14.2

Resource Requirements:

Same requirements as for PHY-01.

Last Modification: March 25, 2009

Discussion:

Reference [1] specifies the general PHY conformance limits for SATA products. This specification includes conformance limits for the TX Frequency Long Term Accuracy. Reference [2] provides the definition of this term for the purposes of SATA testing. Reference [3] defines the measurement requirements for this test. Reference [4] describes the procedure for placing the PUT into BIST mode to generate the required test patterns.

The test is only run on products without SSC enabled. An SSC Track is used to demodulate the signal and apply a low pass filter, as in PHY-01. The formula used for this measurement is the following:

[(mean frequency – nominal frequency) / nominal frequency] * 1e6ppm.

This test is only run once at the maximum interface rate of the product (1.5Gb/s, 3.0Gb/s or 6.0Gb/s).

Test Setup:

Same setup as for PHY-01.

Test Procedure:

The channels should be deskewed before performing this test. See Appendix A for details. The test procedure then proceeds as follows:

1) Open the QualiPHY SATA PHY, TSG, OOB, RSG test suite. See Appendix B for details.
2) Select an appropriate configuration for the maximum supported speed with PHY-02 selected and with the proper SSC setting.
3) Run the test.
4) If using a PeRT3, it will place the product in BIST and cause it to output the required patterns. Otherwise, when prompted to produce HFTP set the product to generate HFTP. See Appendix C for details. Check the pattern on the oscilloscope to make sure it is correct. Press "OK" to continue.
5) When the test completes, generate a report. Observe the results for PHY-02 in the report. This value is reported as "ftol".

Observable Results:

The Frequency Long Term Accuracy value shall be between +/- 350ppm for products running at either 1.5Gb/s, 3.0Gb/s or 6.0Gb/s.

Possible Problems:

See PHY-01.

Test PHY-03 – Spread-Spectrum Modulation Frequency

Purpose: To verify that the Spread Spectrum Modulation Frequency of the PUT's transmitter is within the conformance limits.

References:

[1] SATA Standard, 7.4.2, Table 52 – General Specifications
[2] Ibid, 7.4.3.1.6 – Spread Spectrum Modulation Frequency
[3] Ibid, 7.6.16 – SSC Profile
[4] SATA PRE-TEST MOI
[5] SATA Interoperability Program Unified Test Document, 3.14.3

Resource Requirements:

Same requirements as for PHY-01.

Last Modification: March 25, 2009

Discussion:

Reference [1] specifies the general PHY conformance limits for SATA products. This specification includes conformance limits for the Spread-Spectrum Modulation Frequency. Reference [2] provides the definition of this term for the purposes of SATA testing. Reference [3] defines the measurement requirements for this test. Reference [4] describes the procedure for placing the PUT into BIST mode to generate the required test patterns.

The test is only run on products with SSC enabled. An SSC Track is used to demodulate the signal and apply a low pass filter, as in PHY-01.The Spread-Spectrum Modulation Frequency, f_{SSC}, is measured from the SSC Track.

This test is only run once at the maximum interface rate of the product (1.5Gb/s, 3.0Gb/s or 6.0Gb/s).

Test Setup:

Same setup as for PHY-01.

Test Procedure:

The channels should be deskewed before performing this test. See Appendix A for details. The test procedure then proceeds as follows:

1) Open the QualiPHY SATA PHY, TSG, OOB, RSG test suite. See Appendix B for details.
2) Select an appropriate configuration for the maximum supported speed with PHY-03 selected and with the proper SSC setting.
3) Run the test.
4) If using a PeRT3, it will place the product in BIST and cause it to output the required patterns. Otherwise, when prompted to produce HFTP set the product to generate HFTP. See Appendix C for details. Check the pattern on the oscilloscope to make sure it is correct. Press "OK" to continue.
5) When the test completes, generate a report. Observe the results for PHY-03 in the report. This value is reported as "f_{SSC}".

Observable Results:

The Spread-Spectrum Modulation Frequency value shall be between 30 and 33 kHz for products running at 1.5Gb/s, 3.0Gb/s or 6.0Gb/s.

Possible Problems:

We have seen products that aim for exactly 30 kHz modulation. Since determining f_{SSC} from 10 cycles of a somewhat noisy SSC Track does not have repeatability to five digits, in such a case it is possible for the report to show slightly less than 30 kHz and "FAIL". Products that aim for exactly 30 kHz f_{SSC} give themselves approximately a 50% chance of passing this test. 30 kHz is a lower limit, not a goal; so any observation of a value

below 30 kHz should be considered a failure. Products should leave some margin by aiming for frequency at least 0.1 kHz higher than 30 kHz. Figure 1, below, shows a screen image from a report and Figure 2 shows part of the Summary Table from that report.

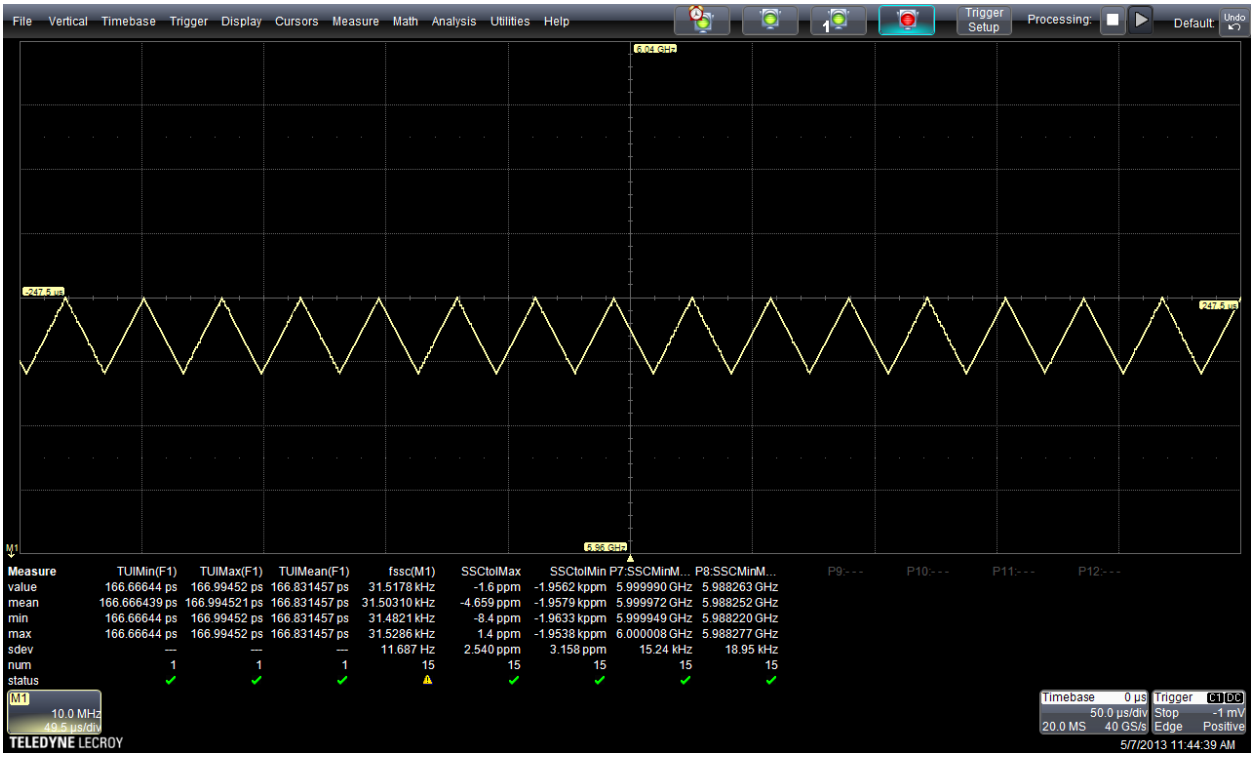


Figure 1. Image of an SSCTrack and measurements for products with SSC

Pass	Run #	Test	Measurement	Current Value	Test Criteria
ℹ	1	PHY-01	Unit Interval Min	166.6664 ps	Informational Only
ℹ	1	PHY-01	Unit Interval Max	166.9945 ps	Informational Only
✔	1	PHY-01	Unit Interval Mean	166.8315 ps	166.6083 ps <= x <= 167.5583 ps
✔	1	PHY-03	fSSC	31.503 kHZ	30.000 kHZ <= x <= 33.000 kHZ
✔	1	PHY-04	SSCtol Max	-5 ppm	-5.350 kppm <= x <= 350 ppm
✔	1	PHY-04	SSCtol Min	-1.958 kppm	-5.350 kppm <= x <= 350 ppm

Figure 2. Summary Table shown in report for SSC measurements

Also see PHY-01.

Test PHY-04 – Spread-Spectrum Modulation Deviation

Purpose: To verify that the Spread-Spectrum Modulation Deviation of the PUT's transmitter is within the conformance limits.

References:

[1] SATA Standard, 7.4.2, Table 52 – General Specifications
[2] Ibid, 7.4.3.1.7 – Spread Spectrum Modulation Deviation
[3] Ibid, 7.6.16 – SSC Profile
[4] SATA PRE-TEST MOI
[5] SATA Interoperability Program Unified Test Document, 3.14.4

Resource Requirements:

Same requirements as for PHY-01.

Last Modification: March 25, 2009

Discussion:

Reference [1] specifies the general PHY conformance limits for SATA products. This specification includes conformance limits for the Spread-Spectrum Modulation Deviation. Reference [2] provides the definition of this term for the purposes of SATA testing. Reference [3] defines the measurement requirements for this test. Reference [4] describes the procedure for placing the PUT into BIST mode to generate the required test patterns. Reference [6] corrects the limits of SSC_{tol} so that they do not conflict with the unit interval limits.

The test is only run on products with SSC enabled. An SSC Track is used to demodulate the signal and apply a low pass filter, as in PHY-01. The Spread-Spectrum Modulation deviation, SSC_{TOL}, is measured from the SSC Track. The results reported are the upper and lower values of the SSC Track calculated as follows:
Upper: ((mean of the max frequency over 10 SSC cycles) – nominal frequency)/nominal frequency * 1e6 ppm.
Lower: ((mean of the min frequency over 10 SSC cycles) – nominal frequency)/nominal frequency * 1e6 ppm.

This test is only run once at the maximum interface rate of the product (1.5Gb/s, 3.0Gb/s or 6.0Gb/s).

Test Setup:

Same setup as for PHY-01.

Test Procedure:

The channels should be deskewed before performing this test. See Appendix A for details. The test procedure then proceeds as follows:

1) Open the QualiPHY SATA PHY, TSG, OOB, RSG test suite. See Appendix B for details.
2) Select an appropriate configuration for the maximum supported speed with PHY-04 selected and with the proper SSC setting.
3) Run the test.
4) If using a PeRT3, it will place the product in BIST and cause it to output the required patterns. Otherwise, when prompted to produce HFTP set the product to generate HFTP. See Appendix C for details. Check the pattern on the oscilloscope to make sure it is correct. Press "OK" to continue.
5) When the test completes, generate a report. Observe the results for PHY-04 in the report. This value is reported as "SSCtol".

Observable Results:

The Spread-Spectrum Modulation Deviation value shall be between –5350ppm and +350ppm for products running at either 1.5Gb/s, 3.0Gb/s or 6.0Gb/s.

Possible Problems:

See PHY-01.

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PHY TRANSMITTED SIGNAL REQUIREMENTS

Overview:

This group of tests verifies the PHY Transmitted Signal Requirements, as defined in Section 2.16 of the SATA Interoperability Unified Test Document (which references the SATA Standard).

Test TSG-01 – Differential Output Voltage

Purpose: To verify that the Differential Output Voltage of the PUT's transmitter is within the conformance limits.

References:

[1] SATA Standard, 7.4.2, Table 54 – Transmitted Signal Requirements
[2] Ibid, 7.4.3.3.2 – TX Differential Output Voltage
[3] Ibid, 7.6.4 – Measurement of Differential Voltage Amplitudes
[4] Ibid, 7.6.7 – Transmitter Amplitude
[5] SATA PRE-TEST MOI
[6] SATA Interoperability Program Unified Test Document, 3.16.2

Resource Requirements:

Same as for PHY-01, repeated here for convenience:
Teledyne LeCroy oscilloscope (see Appendix A for specific models) with 7.2.x.x or later firmware, with QPHY-SATA option key.
Teledyne LeCroy QualiPHY SATA test suite version 7.2.x.x.
SATA test fixture and 2 SMA cables.
BIST Initiator - – One of the following or any other mechanism that makes the product produce the required patterns:
Teledyne LeCroy PeRT3 with SATA option, software version 2.54.406.
Teledyne LeCroy Sierra M6-2 or M6-4 Protocol Test System with software version 3.40 or later.
Teledyne LeCroy SASTrainer Protocol Exerciser with SASTracer software version 2.80 or later.

See appendix A for details.

Last Modification: May 6, 2013

Discussion:

Reference [1] specifies the Transmitted Signal conformance limits for SATA products. This specification includes conformance limits for the Differential Output Voltage. Reference [2] provides the definition of this term for the purposes of SATA testing. References [3] and [4] define the measurement requirements for this test. Reference [5] describes the procedure for placing the PUT into BIST mode to generate the required test patterns.

The QualiPHY SATA test suite acquires one acquisition of each the required test patterns on both ends of the differential pair, and uses those acquisitions for all PHY and TSG tests. The acquisition sizes vary for different patterns based on which tests require a given pattern. For tests that do not require the full acquisition size sometimes only part of the acquisition is used to make the measurement. The acquisition sizes are as follows:

HFTP, LBP, SSOP: 500µs (750,000 UI at 1.5Gb/s, 1.5 million UI at 3.0Gb/s)
MFTP, LFTP: 20µs (30,000 UI at 1.5Gb/s, 60,000 UI for 3.0Gb/s)

The differential output voltage is measured by recovering the clock from the data and slicing the waveform on the recovered clock's UI boundary according to the pattern. A PLL with the same filter characteristics as in TSG-09-12 is used in recovering the clock. The slices are then accumulated into a persistence map. Then a histogram is created from the values of the persistence map over the UI interval specified in [3]. Then VdiffMin and VdiffMax are computed as described in [3].

VdiffMin is tested with HFTP, MFTP and LBP. Figure 3 shows a screenshot of a VdiffMin (HFTP) measurement taken from a test report on a product running at 3.0Gb/s. VdiffMax is tested with MFTP and LFTP. Figure 4 shows a screenshot of a VdiffMax (MFTP) measurement for a product running at 3.0Gb/s. In both, images F2 and F3 show the persistence map of TX+ and TX-, respectively. F6 and F7 show the corresponding histograms. The region over which the histogram was built is highlighted. In Figure 3, P3 shows the differential output voltage "DH". In Figure 4, P3 and P6 show pu and pl, respectively.

For products which support a maximum interface rate of 3.0Gb/s or 6.0Gb/s, this requirement must be tested at both interface rates (1.5Gb/s and 3.0Gb/s).

For products running at 6.0Gb/s refer to Test TSG-14 and Test TSG-15.

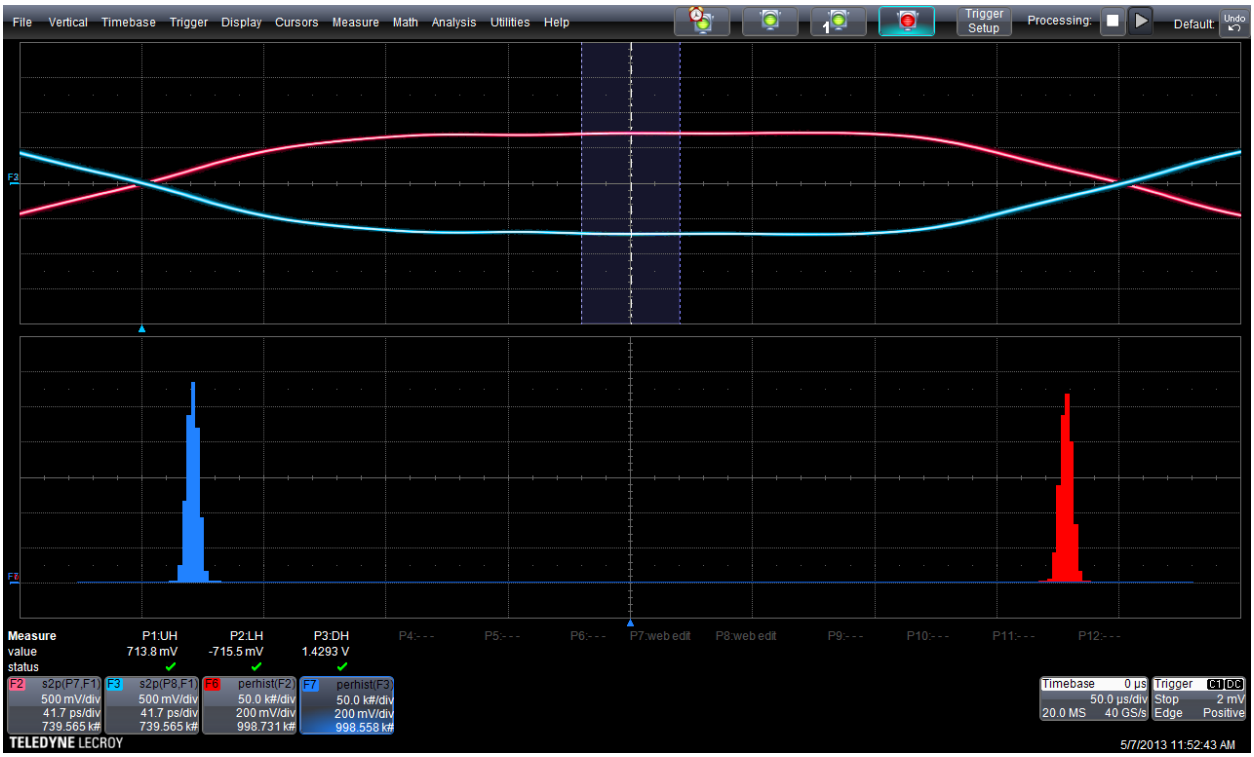


Figure 3. Image of VdiffMin measurement for a product running at 3.0Gb/s transmitting HFTP

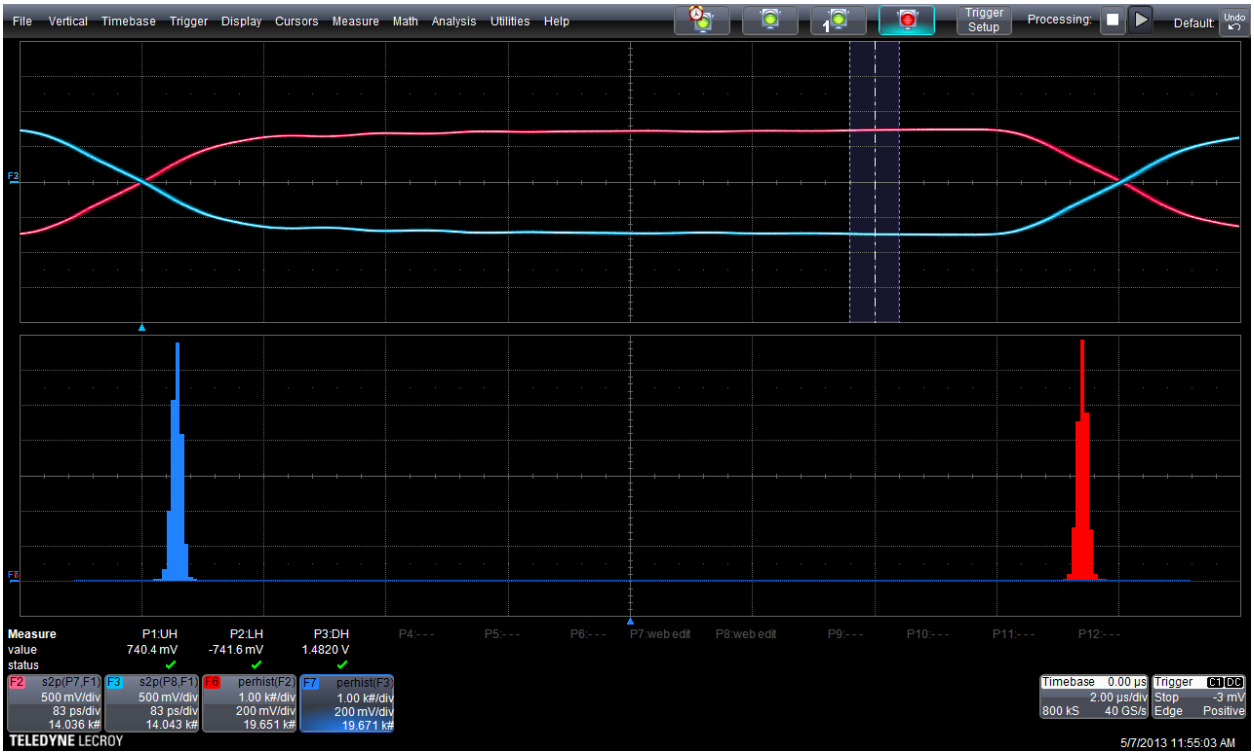


Figure 4. Image of VdiffMax measurement for a product running at 3.0Gb/s transmitting MFTP

Test Setup:

Channels 1 and 2 of the oscilloscope are connected to the transmitter pair on the SATA test fixture. For devices this is 5 and 6, and for hosts it is 2 and 3. The SATA test fixture is inserted into the PUT. For hosts, this test should be done on the worst case port identified. See Reference [5] for details.

Test Procedure:

The channels should be deskewed before performing this test. See Appendix A for details. The test procedure then proceeds as follows:

1) Open the QualiPHY SATA PHY, TSG, OOB, RSG test suite. See Appendix B for details.

For products that support 3.0Gb/s, steps 2-5 must be performed at both 1.5Gb/s and 3.0Gb/s speeds:

2) Select an appropriate configuration with TSG-01 selected.
3) Run the test.
4) If using a PeRT3, it will place the product in BIST and cause it to output the required patterns. Otherwise, when prompted to produce a required test pattern, set the product to generate that pattern. See Appendix C for details. Check the pattern on the oscilloscope to make sure it is correct. Press "OK" to continue.
5) When the test completes, generate a report. Observe the results for TSG-01 in the report. The individual results for VdiffMin with HFTP, MFTP and LBP are reported as DH, DM and VTestLBP, respectively. Each value "passes" if it meets the VdiffMin requirements. The minimum of the three values should be recorded as VdiffMin. TSG-01 passes if VdiffMin passes.

Observable Results:

VdiffMin shall be at least 400 mV. The values of pu and pl are provided only for informational purposes.

Possible Problems: See PHY-01.

Test TSG-02 – Rise/Fall Time (Informative)

Purpose: To verify that the Rise/Fall time of the PUT's transmitter is within the conformance limits.

References:

[1] SATA Standard, 7.4.2, Table 54 – Transmitted Signal Requirements
[2] Ibid, 7.4.3.3.4– TX Rise/Fall Time
[3] Ibid, 7.6.6 – Rise and Fall Times
[4] SATA PRE-TEST MOI
[5] SATA Interoperability Program Unified Test Document, 3.16.3

Resource Requirements:

Same as for TSG-01.

See appendix A for details.

Last Modification: May 6, 2013

Discussion:

Reference [1] specifies the Transmitted Signal conformance limits for SATA products. This specification includes conformance limits for the Rise/Fall Time. Reference [2] provides the definition of this term for the purposes of SATA testing. Reference [3] defines the measurement requirements for this test. Reference [4] describes the procedure for placing the PUT into BIST mode to generate the required test patterns.

TSG-02 is tested using LFTP. Rise and fall measurements are made on the differential signal between the 20% and 80% levels of the waveform's amplitude. In this measurement, amplitude is determined from the statistical mode of the data point values in the waveform. This measurement is made on a 14µs portion of the acquired waveforms described in TSG-01. This amounts to 21,000 UIs at 1.5Gb/s, 42,000 UIs at 3.0Gb/s or 84,000 UIs at 6.0Gb/s.

The cables connecting the SATA test fixture to the oscilloscope must be deskewed. Skew lengthens the measured differential rise and fall times.

For products which support interface rates above 1.5Gb/s, this requirement must be tested at all interface rates (1.5Gb/s, 3.0Gb/s and 6.0Gb/s) for informational purposes only.

Test Setup:

Same setup as for TSG-01.

Test Procedure:

The channels should be deskewed before performing this test. See Appendix A for details. The test procedure then proceeds as follows:

1) Open the QualiPHY SATA PHY, TSG, OOB, RSG test suite. See Appendix B for details.

For products that support speeds greater than 1.5Gb/s, steps 2-5 must be performed at all speeds (1.5Gb/s, 3.0Gb/s and 6.0Gb/s):

2) Select an appropriate configuration with TSG-02 selected.
3) Run the test.
4) If using a PeRT3, it will place the product in BIST and cause it to output the required patterns. Otherwise, when prompted to produce a required test pattern set the product to generate that pattern. See Appendix C for details. Check the pattern on the oscilloscope to make sure it is correct. Press "OK" to continue.
5) When the test completes, generate a report. Observe the results for TSG-02 in the report. The differential rise and fall times are reported, for LFTP. TSG-02 passes only if both the LFTP differential rise and differential fall times pass.

Observable Results:

The TX Rise/Fall Times shall meet the RFT max limit for LFTP specified in Reference [1]. For convenience, the values are reproduced below.

Product Type	RFT Min	RFT Max
1.5Gb/s	50ps	273ps
3.0Gb/s	50ps	136ps
6.0Gb/s	33ps	80ps

Possible Problems:

See PHY-01.

Test TSG-03 – Differential Skew (Informative)

Purpose: To verify that the Differential Skew of the PUT's transmitter is within the conformance limits.

References:

[1] SATA Standard, 7.4.2, Table 54– Transmitted Signal Requirements
[2] Ibid, 7.4.3.3.5 – TX Differential Skew
[3] Ibid, 7.6.17 – Intra-pair Skew
[4] SATA PRE-TEST MOI
[5] SATA Interoperability Program Unified Test Document, 3.16.4

Resource Requirements:

Same as for TSG-01.

See appendix A for details.

Last Modification: May 6, 2013

Discussion:

Reference [1] specifies the Transmitted Signal conformance limits for SATA products. This specification includes conformance limits for Differential Skew. Reference [2] provides the definition of this term for the purposes of SATA testing. Reference [3] defines the measurement requirements for this test. Reference [4] describes the procedure for placing the PUT into BIST mode to generate the required test patterns.

This test is performed by measuring the mean skew of TX+ rise to TX- fall and TX+ fall to TX- rise at the 50% levels of the single-ended waveforms. The 50% level is computed with respect to the waveform's amplitude as determined from the statistical distribution of data point values in the waveform. By using the 50% level the measurement works even in the presence of a DC offset. The measurement is made on a 14μs portion of the acquired waveforms described in TSG-01. This amounts to 21,000 UIs at 1.5Gb/s, 42,000 UIs at 3.0Gb/s or 84,000 UIs at 6.0Gb/s.

The cables connecting the SATA test fixture to the oscilloscope must be deskewed before the data is collected. Uncompensated cable skew contributes directly to measured differential skew.

This test is only run once at the maximum interface rate of the product (1.5Gb/s, 3.0Gb/s or 6.0Gb/s) for informational purposes only.

Test Setup:

Same as for TSG-01.

Test Procedure:

The channels should be deskewed before performing this test. See Appendix A for details. The test procedure then proceeds as follows:

6) Open the QualiPHY SATA PHY, TSG, OOB, RSG test suite. See Appendix B for details.
7) Select an appropriate configuration for the maximum supported speed with TSG-03 selected.
8) Run the test.
9) If using a PeRT3, it will place the product in BIST and cause it to output the required patterns. Otherwise, when prompted to produce a required test pattern set the product to generate that pattern. See Appendix C for details. Check the pattern on the oscilloscope to make sure it is correct. Press "OK" to continue.
10) When the test completes, generate a report. Observe the results for TSG-03 in the report. Differential Skew is reported for HFTP and separately for MFTP. TSG-03 passes only if both results, Differential Skew for HFTP and for MFTP, pass.

Observable Results:

The TX Differential Skew shall be at most 20ps for products running at either 1.5Gb/s, 3.0Gb/s or 6.0Gb/s.

Possible Problems: See PHY-01.

Test TSG-04 – AC Common Mode Voltage

Purpose: To verify that the AC Common Mode Voltage of the PUT's transmitter is within the conformance limits.

References:

[1] SATA Standard, 7.4.2, Table 54 – Transmitted Signal Requirements
[2] Ibid, 7.4.3.3.6 – TX AC Common Mode Voltage (Gen2i, Gen2m)
[3] Ibid, 7.4.3.3.7 – TX AC Common Mode Voltage (Gen1u, Gen2u, Gen3i, Gen3u)
[4] Ibid, 7.6.22 – TX AC Common Mode Voltage (Gen2i, Gen2m)
[5] Ibid, 7.6.23 – TX AC Common Mode Voltage (Gen1u, Gen2u, Gen3i, Gen3u)
[6] SATA PRE-TEST MOI
[7] SATA Interoperability Program Unified Test Document, 3.16.5

Resource Requirements:

Same as for TSG-01.

See appendix A for details.

Last Modification: May 6, 2013

Discussion:

Reference [1] specifies the Transmitted Signal conformance limits for SATA products. This specification includes conformance limits for the TX AC Common Mode Voltage. References [2] and [3] provide the definition of this term for the purposes of SATA testing. References [4] and [5] define the measurement requirements for this test. Reference [6] describes the procedure for placing the PUT into BIST mode to generate the required test patterns.

This test is performed by applying a band pass filter to (TX+ - TX-)/2 with a low frequency cutoff at 200MHz and a high frequency cutoff of half the bit rate as described in References [4] and [5]. The band pass filter should have a first order roll off. Note: Reference [4] clearly defines a measurement that appears to be different than definition of the term in Reference [2]. TSG-04 is tested with a 20µs acquisition of MFTP for Gen1u, Gen2u, Gen2i, Gen2m, Gen3i and Gen3u. TSG-04 is also tested with a 20µs acquisition of HFTP for Gen1u, Gen2u, Gen3i, and Gen 3u.

The cables connecting the SATA test fixture to the oscilloscope must be deskewed before data is captured. Skew contributes directly to common mode spikes which if large enough, even though they are low pass filtered to half the bit rate, can cause failure.

Test Setup:

Same as for TSG-01.

Test Procedure:

The channels should be deskewed before performing this test. See Appendix A for details. The test procedure then proceeds as follows:

1) Open the QualiPHY SATA PHY, TSG, OOB, RSG test suite. See Appendix B for details.
2) Select an appropriate configuration for a product running at 3.0Gb/s with TSG-04 selected.
3) Run the test.
4) If using a PeRT3, it will place the product in BIST and cause it to output the required patterns. Otherwise, when prompted to produce a required test pattern set the product to generate that pattern. See Appendix C for details. Check the pattern on the oscilloscope to make sure it is correct. Press "OK" to continue.
5) When the test completes, generate a report. Observe the results for TSG-04 in the report. The report shows it as Vcm,acTX max, that is the term used in Reference [1]. AC Common Mode is reported for MFTP.

Observable Results:

The AC Common Mode Voltage value shall be at most 50mVpp (Gen2i, Gen2m), 100mVpp (Gen1u, Gen2u) and 120mVpp (Gen3i, Gen3u).

Possible Problems: See PHY-01.

Test TSG-05 – Rise/Fall Imbalance (Obsolete)

Purpose: To verify that the Rise/Fall Imbalance of the PUT's transmitter is within the conformance limits.

References:

[1] SATA Standard, 7.4.2, Table 54 – Transmitted Signal Requirements
[2] Ibid, 7.4.3.3.11 – TX Rise/Fall Imbalance
[3] Ibid, 7.6.21 – TX Rise/Fall Imbalance
[4] SATA PRE-TEST MOI
[5] SATA Interoperability Program Unified Test Document, 3.16.6

Resource Requirements:

Same as for TSG-01.

See appendix A for details.

Last Modification: March 25, 2009

Discussion:

Reference [1] specifies the Transmitted Signal conformance limits for SATA products. This specification includes conformance limits for the Rise/Fall Imbalance. Reference [2] provides the definition of this term for the purposes of SATA testing. Reference [3] defines the measurement requirements for this test. Reference [4] describes the procedure for placing the PUT into BIST mode to generate the required test patterns.

Rise/Fall imbalance is measured with LFTP. References [2] and [3] both define two values to be computed, for each pattern. For each UI the following values are computed:

abs(TX+ rise - TX- fall) / ((TX+ rise + TX- fall) / 2) * 100 %
abs(TX+ fall - TX- rise) / ((TX+ fall + TX- rise) / 2) * 100 %

The results are the means for each computation. The measurement is made on a 14µs portion of the acquired waveforms described in TSG-01. This amounts to 42,000 UIs at 3.0Gb/s.

This test requirement is only applicable to products with a maximum supported rate of 3.0Gb/s.

Test Setup:

Same as for TSG-01.

Test Procedure:

The channels should be deskewed before performing this test. See Appendix A for details. The test procedure then proceeds as follows:

1) Open the QualiPHY SATA PHY, TSG, OOB, RSG test suite. See Appendix B for details.
2) Select an appropriate configuration for a product running at 3.0Gb/s with TSG-05 selected.
3) Run the test.
4) If using a PeRT3, it will place the product in BIST and cause it to output the required patterns. Otherwise, when prompted to produce a required test pattern set the product to generate that pattern. See Appendix C for details. Check the pattern on the oscilloscope to make sure it is correct. Press "OK" to continue.
5) When the test completes, generate a report. Observe the results for TSG-05 in the report. The two values for rise/fall imbalance are reported for LFTP. TSG-05 passes only if both Rise Imbalance and Fall Imbalance for LFTP passes. If the rise or fall imbalance values fail, the single ended rise and fall times, which are listed in the report but are not part of the IL test, may be of interest.

Observable Results:

The Rise/Fall Imbalance values shall be at most 20% for products running at 3.0Gb/s.

Possible Problems: See PHY-01.

Test TSG-06 – Amplitude Imbalance (Obsolete)

Purpose: To verify that the Amplitude Imbalance of the PUT's transmitter is within the conformance limits.

References:

[1] SATA Standard, 7.4.2, Table 54 – Transmitted Signal Requirements
[2] Ibid, 7.4.3.3.12 – TX Amplitude Imbalance (Gen2i, Gen2m, Gen2u, Gen3i, Gen3u)
[3] Ibid, 7.6.20 – TX Amplitude Imbalance
[4] SATA PRE-TEST MOI
[5] SATA Interoperability Program Unified Test Document, 3.16.7

Resource Requirements:

Same as for TSG-01.

See appendix A for details.

Last Modification: March 25, 2009

Discussion:

Reference [1] specifies the Transmitted Signal conformance limits for SATA products. This specification includes conformance limits for the TX Amplitude Imbalance. Reference [2] provides the definition of this term for the purposes of SATA testing. Reference [3] defines the measurement requirements for this test. Reference [4] describes the procedure for placing the PUT into BIST mode to generate the required test patterns.

Amplitude Imbalance is measured with HFTP and MFTP and is computed as follows:

Abs((TX+ amplitude - TX- amplitude) / ((TX+ amplitude – TX- amplitude) / 2)) * 100 %

In this measurement, amplitude is determined from the mode of the statistical distribution of data point values in the center of the UI. For MFTP, the measurement is made on the second bit. The measurement is made on a 14μs portion of the acquired waveforms described in TSG-01. This amounts to 42,000 UIs at 3.0Gb/s.

This test requirement is only applicable to products with a maximum supported rate of 3.0Gb/s.

Test Setup:

Same as for TSG-01.

Test Procedure:

The channels should be deskewed before performing this test. See Appendix A for details. The test procedure then proceeds as follows:

1) Open the QualiPHY SATA PHY, TSG, OOB, RSG test suite. See Appendix B for details.
2) Select an appropriate configuration for a product running at 3.0Gb/s with TSG-06 selected.
3) Run the test.
4) If using a PeRT3, it will place the product in BIST and cause it to output the required patterns. Otherwise, when prompted to produce a required test pattern set the product to generate that pattern. See Appendix C for details. Check the pattern on the oscilloscope to make sure it is correct. Press "OK" to continue.
5) When the test completes, generate a report. Observe the results for TSG-06 in the report. The values for amplitude imbalance are reported for HFTP and separately for MFTP. TSG-06 passes only if AmpBal passes for both HFTP and for MFTP.

Observable Results:

The TX Amplitude Imbalance value shall be at most 10% for products running at 3.0Gb/s.

Possible Problems:

See PHY-01.

Test TSG-07 – Gen1 (1.5Gb/s) TJ at Connector, Clock to Data, f_{BAUD} /10 (Obsolete)

This measurement is no longer defined in Serial ATA Revision 3.2 and later and has been removed.

Test TSG-08 – Gen1 (1.5Gb/s) DJ at Connector, Clock to Data, $f_{BAUD}/10$ (Obsolete)

This measurement is no longer defined in Serial ATA Revision 3.2 and later and has been removed.

Test TSG-09 – Gen1 (1.5Gb/s) TJ at Connector, Clock to Data, f_{BAUD} /500 JTF Defined

Purpose: To verify that the Gen1 (1.5Gb/s) TJ at Connector (Clock to Data, $f_{BAUD}/500$) of the PUT's transmitter is within the conformance limits.

References:

[1] SATA Standard, 7.4.2, Table 54 – Transmitted Signal Requirements
[2] Ibid, 7.4.3.3.11 – Clock-to-Data Transmit Jitter (Gen1i, Gen1m, Gen1u, Gen2i, Gen2m, Gen2u, Gen3i, Gen3u)
[3] Ibid, 7.5.3 – Reference Clock Definition
[4] Ibid, 7.6.10 – Jitter Measurements
[5] SATA Interoperability Program Unified Test Document, 3.16.10
[6] SATA PRE-TEST MOI

Resource Requirements:

Same as for TSG-01.

See appendix A for details.

Last Modification: March 25, 2009

Discussion:

Reference [1] specifies the Transmitted Signal conformance limits for SATA products. Reference [2] provides the definition of this term for the purposes of SATA testing. Reference [3] defines the measurement requirements for this test. Reference [6] describes the procedure for placing the PUT into BIST mode to generate the required test patterns.

The Unified Test document specifies that this test should be made with HFTP and LBP, and if time permits should also be made with SSOP. The test is performed by making an edge to reference jitter measurement. A PLL meeting the Jitter Transfer Function requirements in [3] is used to recover the clock. Follow the procedure in Appendix E to set the PLL appropriately. A 500µs acquisition is used for this test corresponding to 750,000 UI, as described in TSG-01.

For products which support interface rates above 1.5Gb/s, this requirement must be tested at 1.5Gb/s.

Test Setup:

Same as for TSG-01.

Test Procedure:

The channels should be deskewed before performing this test. See Appendix A for details. The test procedure then proceeds as follows:

1) Open the QualiPHY SATA PHY, TSG, OOB, RSG test suite. See Appendix B for details.
2) Select an appropriate configuration for a product running at 1.5Gb/s with TSG-09 selected and the PLL natural frequency and damping set correctly (see Appendix E).
3) Run the test.
4) If using a PeRT3, it will place the product in BIST and cause it to output the required patterns. Otherwise, when prompted to produce a required test pattern set the product to generate that pattern. See Appendix C for details. Check the pattern on the oscilloscope to make sure it is correct. Press "OK" to continue.
5) When the test completes, generate a report. Observe the results for TSG-09 in the report. The values for TJ, $f_{BAUD}/500$ are reported for HFTP and separately for LBP and SSOP. TSG-09 passes only if both the HFTP and LBP values pass.

Observable Results:

The TJ shall be at most 0.37UI when measured at $f_{BAUD}/500$ for products running at 1.5Gb/s.

Possible Problems: See PHY-01.

Test TSG-10 – Gen1 (1.5Gb/s) DJ at Connector, Clock to Data, f_{BAUD}/500 JTF Defined

Purpose: To verify that the Gen1 (1.5Gb/s) DJ at Connector (Clock to Data, f_{BAUD}/500) of the PUT's transmitter is within the conformance limits.

References:

[1] SATA Standard, 7.4.2, Table 54 – Transmitted Signal Requirements
[2] Ibid, 7.4.3.3.11 – Clock-to-Data Transmit Jitter (Gen1i, Gen1m, Gen1u, Gen2i, Gen2m, Gen2u, Gen3i, Gen3u)
[3] Ibid, 7.5.3 – Reference Clock Definition
[4] Ibid, 7.6.10 – Jitter Measurements
[5] SATA Interoperability Program Unified Test Document, 3.16.11
[6] SATA PRE-TEST MOI

Resource Requirements:

Same as for TSG-01.

See appendix A for details.

Last Modification: March 25, 2009

Discussion:

Reference [1] specifies the Transmitted Signal conformance limits for SATA products. Reference [2] provides the definition of this term for the purposes of SATA testing. Reference [3] defines the measurement requirements for this test. Reference [6] describes the procedure for placing the PUT into BIST mode to generate the required test patterns.

The Unified Test document specifies that this test should be made with HFTP and LBP, and if time permits should also be made with SSOP. The test is performed by making an edge to reference jitter measurement. A PLL meeting the Jitter Transfer Function requirements in [3] is used to recover the clock. Follow the procedure in Appendix E to set the PLL appropriately. A 500µs acquisition is used for this test corresponding to 750,000 UI, as described in TSG-01.

For products which support interface rates above 1.5Gb/s, this requirement must be tested at 1.5Gb/s.

Test Setup:

Same as for TSG-01.

Test Procedure:

The channels should be deskewed before performing this test. See Appendix A for details. The test procedure then proceeds as follows:

1) Open the QualiPHY SATA PHY, TSG, OOB, RSG test suite. See Appendix B for details.
2) Select an appropriate configuration for a product running at 1.5Gb/s with TSG-10 selected and the PLL natural frequency and damping set correctly (see Appendix E).
3) Run the test.
4) If using a PeRT3, it will place the product in BIST and cause it to output the required patterns. Otherwise, when prompted to produce a required test pattern set the product to generate that pattern. See Appendix C for details. Check the pattern on the oscilloscope to make sure it is correct. Press "OK" to continue.
5) When the test completes, generate a report. Observe the results for TSG-10 in the report. The values for DJ, f_{BAUD}/500 are reported for HFTP and separately for LBP and SSOP. TSG-10 passes only if both the HFTP and LBP values pass.

Observable Results:

The DJ shall be at most 0.19UI when measured at f_{BAUD}/500 for products running at 1.5Gb/s.

Possible Problems: See PHY-01.

Test TSG-11 – Gen2 (3.0Gb/s) TJ at Connector, Clock to Data, $f_{BAUD}/500$ JTF Defined

Purpose: verify that the Gen2 (3.0Gb/s) TJ at Connector (Clock to Data, $f_{BAUD}/500$) of the PUT's transmitter is within the conformance limits.

References:

[1] SATA Standard, 7.4.2, Table 54 – Transmitted Signal Requirements
[2] Ibid, 7.4.3.3.11 – Clock-to-Data Transmit Jitter (Gen1i, Gen1m, Gen1u, Gen2i, Gen2m, Gen2u, Gen3i, Gen3u)
[3] Ibid, 7.5.3 – Reference Clock Definition
[4] Ibid, 7.6.10 – Jitter Measurements
[5] SATA PRE-TEST MOI
[6] SATA Interoperability Program Unified Test Document, 3.16.12

Resource Requirements:

Same as for TSG-01.

See appendix A for details.

Last Modification: March 25, 2009

Discussion:

Reference [1] specifies the Transmitted Signal conformance limits for SATA products. This specification includes conformance limits for the TJ at Connector (Clock to Data, $f_{BAUD}/500$). Reference [2] provides the definition of this term for the purposes of SATA testing. Reference [3] defines the measurement requirements for this test. Reference [5] describes the procedure for placing the PUT into BIST mode to generate the required test patterns.

The Unified Test document specifies that this test should be made with HFTP and LBP, and if time permits should also be made with SSOP. The test is performed by making an edge to reference jitter measurement. A PLL meeting the Jitter Transfer Function requirements in [3] is used to recover the clock. Follow the procedure in Appendix E to set the PLL appropriately. A 500µs acquisition is used for this test corresponding to 1.5 million UI, as described in TSG-01.

This test requirement is only applicable to products running at 3.0Gb/s.

Test Setup:

Same as for TSG-01.

Test Procedure:

The channels should be deskewed before performing this test. See Appendix A for details. The test procedure then proceeds as follows:

1) Open the QualiPHY SATA PHY, TSG, OOB, RSG test suite. See Appendix B for details.
2) Select an appropriate configuration for a product running at 3.0Gb/s with TSG-11 selected and the PLL natural frequency and damping set correctly (see Appendix E).
3) Run the test.
4) If using a PeRT3, it will place the product in BIST and cause it to output the required patterns. Otherwise, when prompted to produce a required test pattern set the product to generate that pattern. See Appendix C for details. Check the pattern on the oscilloscope to make sure it is correct. Press "OK" to continue.
5) When the test completes, generate a report. Observe the results for TSG-11 in the report. The values for TJ, $f_{BAUD}/500$ are reported for HFTP and separately for LBP and SSOP. TSG-11 passes only if both the HFTP and LBP values pass.

Observable Results:

The TJ shall be at most 0.37UI when measured at $f_{BAUD}/500$ for products running at 3.0Gb/s.

Possible Problems: See PHY-01.

Test TSG-12 – Gen2 (3.0Gb/s) DJ at Connector, Clock to Data, $f_{BAUD}/500$ JTF Defined

Purpose: To verify that the Gen2 (3.0Gb/s) DJ at Connector (Clock to Data, $f_{BAUD}/500$) of the PUT's transmitter is within the conformance limits.

References:

[1] SATA Standard, 7.4.2, Table 54 – Transmitted Signal Requirements
[2] Ibid, 7.4.3.3.11 – Clock-to-Data Transmit Jitter (Gen1i, Gen1m, Gen1u, Gen2i, Gen2m, Gen2u, Gen3i, Gen3u)
[3] Ibid, 7.5.3 – Reference Clock Definition
[4] Ibid, 7.6.10 – Jitter Measurements
[5] SATA PRE-TEST MOI
[6] SATA Interoperability Program Unified Test Document, 3.16.13

Resource Requirements:

Same as for TSG-01.

See appendix A for details.

Last Modification: March 25, 2009

Discussion:

Reference [1] specifies the Transmitted Signal conformance limits for SATA products. This specification includes conformance limits for the DJ at Connector (Clock to Data, $f_{BAUD}/500$). Reference [2] provides the definition of this term for the purposes of SATA testing. Reference [3] defines the measurement requirements for this test. Reference [5] describes the procedure for placing the PUT into BIST mode to generate the required test patterns.

The Unified Test document specifies that this test should be made with HFTP and LBP, and if time permits should also be made with SSOP. The test is performed by making an edge to reference jitter measurement. A PLL meeting the Jitter Transfer Function requirements in [3] is used to recover the clock. Follow the procedure in Appendix E to set the PLL appropriately. A 500μs acquisition is used for this test corresponding to 1.5 million UI, as described in TSG-01.

This test requirement is only applicable to products running at 3.0Gb/s.

Test Setup:

Same as for TSG-01.

Test Procedure:

The channels should be deskewed before performing this test. See Appendix A for details. The test procedure then proceeds as follows:

1) Open the QualiPHY SATA PHY, TSG, OOB, RSG test suite. See Appendix B for details.
2) Select an appropriate configuration for a product running at 3.0Gb/s with TSG-12 selected and the PLL natural frequency and damping set correctly (see Appendix E).
3) Run the test.
4) If using a PeRT3, it will place the product in BIST and cause it to output the required patterns. Otherwise, when prompted to produce a required test pattern set the product to generate that pattern. See Appendix C for details. Check the pattern on the oscilloscope to make sure it is correct. Press "OK" to continue.
5) When the test completes, generate a report. Observe the results for TSG-12 in the report. The values for DJ, $f_{BAUD}/500$ are reported for HFTP and separately for LBP and SSOP. TSG-12 passes only if both the HFTP and LBP values pass.

Observable Results:

The DJ shall be at most 0.19UI when measured at $f_{BAUD}/500$ for products running at 3.0Gb/s.

Possible Problems: See PHY-01.

Test TSG-13 - Gen3 (6.0Gb/s) Transmit Jitter

Purpose: To verify that the Gen3 (6.0Gb/s) transmit jitter of the PUT is within the conformance limits.

References:

[1] SATA Standard, 7.4.2, Table 54 – Transmitted Signal Requirements
[2] Ibid, 7.4.3.3.11 – Clock-to-Data Transmit Jitter (Gen1i, Gen1m, Gen1u, Gen2i, Gen2m, Gen2u, Gen3i, Gen3u)
[3] Ibid, 7.5.3.4 – Gen3i and Gen3u Normative Requirements
[4] Ibid, 7.6.12 – Transmit Jitter (Gen3i, Gen3u)
[5] SATA PRE-TEST MOI
[6] SATA Interoperability Program Unified Test Document, 3.16.14

Resource Requirements:

Same as for TSG-01.

See appendix A for details.

Last Modification: May 6, 2013

Discussion:

Reference [1] specifies the Transmitted Signal conformance limits for SATA products. This specification includes conformance limits for the RJ measured before the Compliance Interconnect Channel (CIC) and for TJ before and after the CIC. Reference [2] provides the definition of clock to data jitter for the purposes of SATA testing. Reference [3] defines the measurement requirements for this test. Reference [5] describes the procedure for placing the PUT into BIST mode to generate the required test patterns.

For this test the total jitter (TJ) at BER 10E-6 and 10E-12 is measured using an LBP pattern. Additionally HFTP, MFTP, LFTP, and SSOP patterns are tested as informational. For Gen3i PUTs, TJ is measured after the Compliance Interconnect Channel (CIC) for all patterns. For the purposes of this test the CIC is emulated using supplied S-Parameters and the oscilloscope is always connected directly to the PUT using the test fixture. For a Gen3u UHost PUT the Gen3i CIC channel is not used and the measurement is made directly into the lab load. A PLL meeting the Jitter Transfer Function requirements in [3] is used to recover the clock. Follow the procedure in Appendix E to set the PLL appropriately. A 500µs acquisition is used for each pattern tested. This corresponds to 3 million UI at 6.0Gb/s.

This test requirement is only applicable to products running at 6.0Gb/s.

Test Setup:

Same as for TSG-01.

Test Procedure:

The channels should be deskewed before performing this test. See Appendix A for details. The test procedure then proceeds as follows:

1) Open the QualiPHY SATA PHY, TSG, OOB, RSG test suite. See Appendix B for details.
2) Select an appropriate configuration for a product running at 6.0Gb/s with TSG-13 selected and the PLL natural frequency and damping set correctly (see Appendix E).
3) Run the test.
4) If using a PeRT3, it will place the product in BIST and cause it to output the required patterns. Otherwise, when prompted to produce a required test pattern set the product to generate that pattern. See Appendix C for details. Check the pattern on the oscilloscope to make sure it is correct. Press "OK" to continue.
5) When the test completes, generate a report. Observe the results for TSG-13 in the report. The values for TJ after the CIC are reported.

Observable Results:

- Tj values at 10E-6 and 10E-12 measured at a maximum of 0.46UI and 0.52UI respectively into a Laboratory Load after CIC (for Gen3i) or without the CIC (for Gen3u) when measured using the specified JTF (for products running at 6Gb/s)

Possible Problems: See PHY-01.

Test TSG-14 - Gen3 (6.0Gb/s) TX Maximum Differential Voltage Amplitude

Purpose: To verify that the Gen3 (6.0Gb/s) maximum differential voltage of the PUT's transmitter is within the conformance limits.

References:

[1] SATA Standard, 7.4.2, Table 54 – Transmitted Signal Requirements
[2] Ibid, 7.4.3.3.2 – TX Differential Output Voltage
[3] Ibid, 7.6.5.2 – Maximum Differential Voltage Amplitude (Gen3i, Gen3u)
[4] SATA PRE-TEST MOI
[5] SATA Interoperability Program Unified Test Document, 3.16.15

Resource Requirements:

Same as for TSG-01.

See appendix A for details.

Last Modification: March 25, 2009

Discussion:

Reference [1] specifies the Transmitted Signal conformance limits for SATA products. Reference [2] provides the definition this measurement. Reference [3] defines the measurement requirements for this test. Reference [4] describes the procedure for placing the PUT into BIST mode to generate the required test patterns.

The maximum differential voltage amplitude at 6.0Gb/s is measured as the peak to peak voltage after averaging. The measurement is made over a 4 UI span with at least 500 averages. For this test an MFTP pattern is used.

This test requirement is only applicable to products running at 6.0Gb/s.

Test Setup:

Same as for TSG-01.

Test Procedure:

The channels should be deskewed before performing this test. See Appendix A for details. The test procedure then proceeds as follows:

1) Open the QualiPHY SATA PHY, TSG, OOB, RSG test suite. See Appendix B for details.
2) Select an appropriate configuration for a product running at 6.0Gb/s with TSG-14.
3) Run the test.
4) If using a PeRT3, it will place the product in BIST and cause it to output the required patterns. Otherwise, when prompted to produce a required test pattern set the product to generate that pattern. See Appendix C for details. Check the pattern on the oscilloscope to make sure it is correct. Press "OK" to continue.
5) When the test completes, generate a report. Observe the results for TSG-14 VDiffTXMax in the report.

Observable Results:

The maximum differential voltage VDiffTXMax shall be at most 900mV.

Possible Problems: See PHY-01.

Test TSG-15 - Gen3 (6.0Gb/s) TX Minimum Differential Voltage Amplitude

Purpose: To verify that the Gen3 (6.0Gb/s) minimum differential voltage of the PUT's transmitter is within the conformance limits.

References:

[1] SATA Standard, 7.4.2, Table 54 – Transmitted Signal Requirements
[2] Ibid, 7.4.3.3.2 – TX Differential Output Voltage
[3] Ibid, 7.5.3.4 – Gen3i and Gen3u Normative Requirements
[4] Ibid, 7.6.5.3 – Minimum Differential Voltage Amplitude (Gen3i, Gen3u)
[5] SATA PRE-TEST MOI
[6] SATA Interoperability Program Unified Test Document, 3.16.16

Resource Requirements:

Same as for TSG-01.

See appendix A for details.

Last Modification: May 6, 2013

Discussion:

Reference [1] specifies the Transmitted Signal conformance limits for SATA products. Reference [2] provides the definition of this test. Reference [4] defines the measurement requirements for this test. Reference [5] describes the procedure for placing the PUT into BIST mode to generate the required test patterns.

The minimum differential voltage amplitude at 6.0Gb/s is a measurement of the eye opening of an eye diagram between 0.45UI and 0.55UI after the Gen3i Compliance Interconnect Channel (CIC) with a population of at least 5e6 UIs. For the purposes of this test the CIC is emulated using supplied S-Parameters and the oscilloscope is always connected directly to the PUT using the test fixture. For a Gen3u UHost PUT the Gen3i CIC channel is not used and the measurement is made directly into the lab load. In building the eye diagram a PLL meeting the Jitter Transfer Function requirements in [3] is used to recover the clock. Follow the procedure in Appendix E to set the PLL appropriately. For this test an LBP pattern is used. A 1ms acquisition is used to build the eye diagram.

This test requirement is only applicable to products running at 6.0Gb/s.

Test Setup:

Same as for TSG-01.

Test Procedure:

The channels should be deskewed before performing this test. See Appendix A for details. The test procedure then proceeds as follows:

1) Open the QualiPHY SATA PHY, TSG, OOB, RSG test suite. See Appendix B for details.
2) Select an appropriate configuration for a product running at 6.0Gb/s with TSG-15 selected and the PLL natural frequency and damping set correctly (see Appendix E).
3) Run the test.
4) If using a PeRT3, it will place the product in BIST and cause it to output the required patterns. Otherwise, when prompted to produce a required test pattern set the product to generate that pattern. See Appendix C for details. Check the pattern on the oscilloscope to make sure it is correct. Press "OK" to continue.
5) When the test completes, generate a report. Observe the results for TSG-15 VDiffTXMin in the report.

Observable Results:

The minimum differential voltage VDiffTXMin shall be at least 240mV for a device and 200mV for a host.

Possible Problems: See PHY-01.

Test TSG-16 - Gen3 (6.0Gb/s) TX AC Common Mode Voltage (Obsolete)

Purpose: To verify that the Gen3 (6.0Gb/s) AC common mode voltage of the PUT's transmitter is within the conformance limits.

References:

[1] SATA Standard, 7.4.2, Table 54 – Transmitted Signal Requirements
[2] Ibid, 7.4.3.3.7 – TX AC Common Mode Voltage (Gen1u, Gen2u, Gen3i, Gen3u)
[3] Ibid, 7.6.23 – TX AC Common Mode Voltage (Gen1u, Gen2u, Gen3i, Gen3u)
[4] SATA PRE-TEST MOI
[5] SATA Interoperability Program Unified Test Document, 3.16.17

Resource Requirements:

Same as for TSG-01.

See appendix A for details.

Last Modification: May 6, 2013

Discussion:

Reference [1] specifies the Transmitted Signal conformance limits for SATA products. Reference [2] provides the definition of this test. Reference [3] defines the measurement requirements for this test. Reference [4] describes the procedure for placing the PUT into BIST mode to generate the required test patterns.

The AC common mode voltage for a 6.0Gb/s product is measured in the frequency domain using methods equivalent to a spectrum analyzer. The common mode signal is created by summing TX+ and TX- and dividing by two. An FFT with 1MHz resolution is used to measure the signal power at the first and second harmonics. For this test an HFTP pattern is used.

The cables connecting the SATA test fixture to the oscilloscope must be deskewed before data is captured. Skew contributes directly to common mode and can cause failure.

This test requirement is only applicable to products running at 6.0Gb/s.

Test Setup:

Same as for TSG-01.

Test Procedure:

The channels should be deskewed before performing this test. See Appendix A for details. The test procedure then proceeds as follows:

1) Open the QualiPHY SATA PHY, TSG, OOB, RSG test suite. See Appendix B for details.
2) Select an appropriate configuration for a product running at 6.0Gb/s with TSG-16 selected and the PLL natural frequency and damping set correctly (see Appendix E).
3) Run the test.
4) If using a PeRT3, it will place the product in BIST and cause it to output the required patterns. Otherwise, when prompted to produce a required test pattern set the product to generate that pattern. See Appendix C for details. Check the pattern on the oscilloscope to make sure it is correct. Press "OK" to continue.
5) When the test completes, generate a report. Observe the results for TSG-16 VcmacTX fundamental and VcmacTX 2nd harmonic in the report.

Observable Results:

The Transmitter shall not deliver more output voltage than the following limits:

- Fundamental (3 GHz): Max = 26 dBmV(pk)
- 2nd Harmonic (6 GHz): Max = 30 dBmV(pk)

Possible Problems: See PHY-01.

Test TSG-17 - Gen3 (6.0Gb/s) TX Emphasis

Purpose: To verify that the Gen3 (6.0Gb/s) emphasis levels of the PUT's transmitter is within the conformance limits.

References:

[1] SATA Standard, TPR_059 Emphasis Control for SATA Interface
[2] SATA Standard, TPR_069 Emphasis Control for SATA Interface
[3] SATA PRE-TEST MOI
[4] SATA Interoperability Program Unified Test Document, 3.16.18

Resource Requirements:

Same as for TSG-01.

See appendix A for details.

Last Modification: May 1, 2015

Discussion:

References [1] and [2] specify the Transmitted Signal conformance limits for SATA products and provide the definition of this test. References [1] and [2] also define the measurement requirements for this test. Reference [3] describes the procedure for placing the PUT into BIST mode to generate the required test patterns.

There are two different methods to measure the Transmitter emphasis: TPR_059 and TP_069. In TPR_059 the Transmitter emphasis is measured by comparing the HFTP versus the mean differential voltage of the fourth bit of LFTP. According to TPR_069 Transmitter emphasis is measured by comparing the mean differential voltage of the first bit of MFTP versus the mean differential voltage of the second bit of MFTP. In both methods bits are sampled between 0.45 and 0.55 of their respective UIs.

This test requirement is only applicable to products running at 6.0Gb/s.

Test Setup:

Same as for TSG-01.

Test Procedure:

The channels should be deskewed before performing this test. See Appendix A for details. The test procedure then proceeds as follows:

1) Open the QualiPHY SATA PHY, TSG, OOB, RSG test suite. See Appendix B for details.
2) Select an appropriate configuration for a product running at 6.0Gb/s with TSG-17 selected.
3) Run the test.
4) If using a PeRT3, it will place the product in BIST and cause it to output the required patterns. Otherwise, when prompted to produce a required test pattern set the product to generate that pattern. See Appendix C for details. Check the pattern on the oscilloscope to make sure it is correct. Press "OK" to continue.
5) When the test completes, generate a report. Observe the results for TSG-17a (TPR_059) and TSG-17b (TPR_069) in the report.

Observable Results:

- A Gen3 device shall deliver between 0.5 dB and 2.5 dB of Transmitter emphasis.
  - 2 dB is a nominal value as noted in TPR_059.
- A Gen3 host shall deliver between -2 dB and 1.5 dB of Transmitter emphasis.
  - 0 dB is a nominal value as noted in TPR_059.

Possible Problems: See PHY-01.

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PHY OOB REQUIREMENTS

Overview:

This group of tests verifies the Phy OOB Requirements, as defined in Section 3.19 of the SATA Interoperability Unified Test Document (which references the SATA Standard).

Test OOB-01 – OOB Signal Detection Threshold

Purpose: To verify that the OOB Signal Detection Threshold of the PUT's receiver is within the conformance limits.

References:

[1] SATA Standard, 7.4.2, Table 59 – OOB Specifications
[2] Ibid, 7.4.3.7.2 – OOB Signal Detection Threshold
[3] Ibid, 7.6.26 – Squelch Detector Tests
[4] SATA PRE-TEST MOI
[5] SATA Interoperability Program Unified Test Document, 3.19.2

Resource Requirements:

Teledyne LeCroy oscilloscope (see Appendix A for specific models) with 7.2.x.x or later firmware, with QPHY-SATA option key.
Teledyne LeCroy QualiPHY SATA test suite version 7.2.x.x.
SATA test fixture and 2 SMA cables.
Teledyne LeCroy PeRT3 with SATA option, software version 2.54.406
Or
Arbitrary waveform generator (AWG), or equivalent – used to send OOB signals to the PUT.

See appendix A for details.

Last Modification: May 6, 2013

Discussion:

Reference [1] specifies the Transmitted Signal conformance limits for SATA products. This specification includes conformance limits for the OOB Signal Detection Threshold. Reference [2] provides the definition of this term for the purposes of SATA testing. Reference [3] defines the measurement requirements for this test. Reference [4] describes the procedure for placing the PUT into BIST mode to generate the required test patterns.

This test is done by sending COMINIT/COMRESET to the PUT and adjusting the signal amplitude to verify the OOB signal detection threshold of the PUT is within conformance limits. Two signal levels are used, one to verify the PUT does not detect a low amplitude input below the minimum allowable limit and the other to verify the PUT does detect a high amplitude signal above the minimum required limit. When a device detects COMRESET it should respond with COMINIT. Devices that support asynchronous recovery may send unsolicited COMINITs that are not in response to COMRESET. When a host detects COMINIT it should respond with COMWAKE or if asynchronous signal recovery is supported it may respond with COMRESET.

This test is only run once regardless of the maximum interface rate of the product (1.5Gb/s, 3.0Gb/s or 6.0Gb/s).

Test Setup:

If using a PeRT3, set "Use PeRT3" to "Yes" in the setup tab after selecting a configuration. QualiPHY will prompt for the required connections when appropriate and automate the OOB tests using the PeRT3. See Appendix B for more details.

If using an arbitrary waveform generator (AWG), preload the AWG with waveform files to generate COMINIT with nominal OOB Gap timings, see Appendix C for details. Load the COMINIT waveform into the AWG. Use SMA cables to connect the AWG output to the oscilloscope. Follow the procedure in Appendix F to find the vertical amplitude setting required to achieve 210mVppd and 40mVppd (1.5Gb/s) / 60mVppd (3.0Gb/s) at the ends of the SMAs cables. Record these values for use during the test. Connect the SMA cables from AWG to the PUT's receive pair on the fixture. For the Teledyne LeCroy TF-SATA-C fixture use B+ & B- for devices or A+ & A- for hosts. Connect the other pair from the test fixture to C1 and C2 on the oscilloscope. Connect the AWG marker/trigger output to C4 on the oscilloscope.

For hosts, this test should be done on the worst case port identified. See Reference [4] for details.

Test Procedure:

1) Open the QualiPHY SATA PHY, TSG, OOB, RSG test suite. See Appendix B for details.
2) Select an appropriate configuration for the maximum supported speed with with OOB-01 selected.
3) Run the test. If using a PeRT3 the test is executed automatically. If using an AWG continue below:
4) When the prompt appears to run OOB-01 with the 210mV output, load the COMINIT/COMRESET pattern and make the connections described above in Test Setup. Set the output of the AWG to 210mVppd.
5) The oscilloscope should be set to trigger on C4, the AWG marker output, and acquire at least a 2 ms acquisition. The response from the PUT will appear on C1 and C2. Verify that for every COMINIT sent by the AWG there is a response from the PUT. Devices should respond with COMINIT and hosts should respond with COMWAKE or COMRESET. The responses should appear in regular periodic intervals as defined by the output period of the AWG waveform. There should be a response to COMINIT/COMRESET from the PUT following every marker output as shown below in Figure 5. If the product always responds to COMINIT select "Detect" in the message box. Otherwise, select "No Detect."

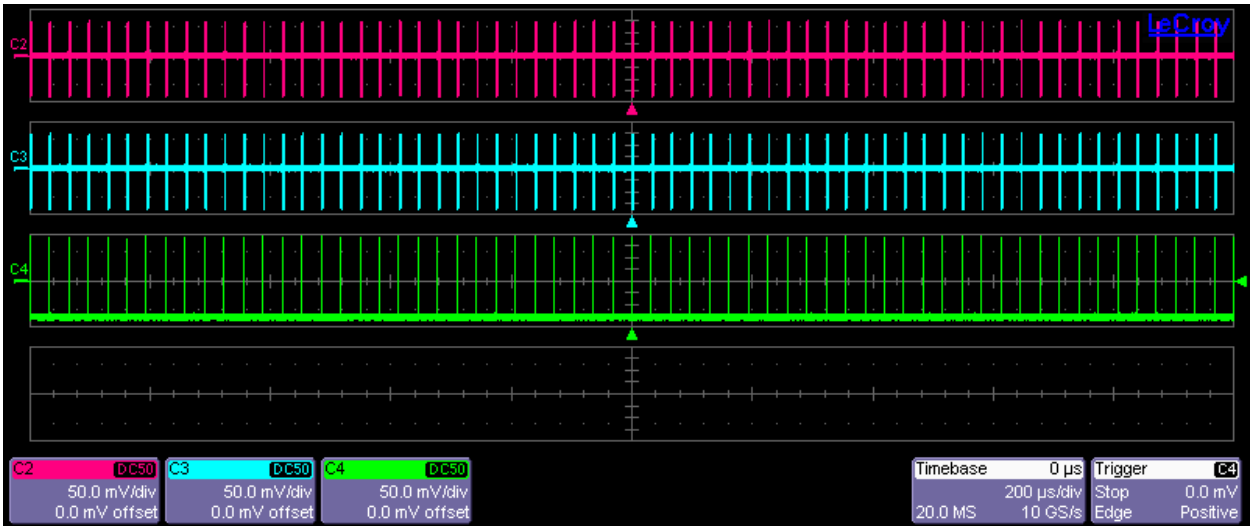


Figure 5. Image of Test OOB-01 with 210mVppd threshold showing a device that detects COMRESET

6) When the prompt appears to run the test with the low threshold set the AWG output to 40mVppd for products that support 1.5Gb/s or 60mVppd for products that support 3.0Gb/s.
7) There should be no COMINIT response from the PUT. If the PUT does not respond to COMINIT/COMRESET select "No Detect" in the message box. Otherwise, select "Detect" in the message box. Note, products that support Asynchronous Signal Recovery may send out unsolicited COMINIT/COMRESET not in response to the COMINIT/COMESET from the AWG. This should be considered a "No Detect."

Observable Results:

Products that support 1.5Gb/s shall not detect OOB at 40mVppd and shall detect OOB at 210mVppd.
Products that support 3.0Gb/s or 6.0Gb/s shall not detect OOB at 60mVppd and shall detect OOB at 210mVppd.

Possible Problems:

Test OOB-02 – UI During OOB Signaling

Purpose: To verify that the UI During OOB Signaling of the PUT's transmitter is within the conformance limits.

References:

[1] SATA Standard, 7.4.2, Table 59 – OOB Specifications
[2] Ibid, 7.4.3.7.3 – UI During OOB Signaling (UIOOB)
[3] Ibid, 7.6.16 – SSC Profile
[4] SATA PRE-TEST MOI
[5] SATA Interoperability Program Unified Test Document, 3.19.3

Resource Requirements:

Same as OOB-01.

See Appendix A for details.

Last Modification: April 29, 2011

Discussion:

Reference [1] specifies the Transmitted Signal conformance limits for SATA products. This specification includes conformance limits for the UI During OOB Signaling. Reference [2] provides the definition of this term for the purposes of SATA testing. Reference [3] defines the measurement requirements for this test.

This test is run by sending the PUT nominal OOB signals and acquiring COMINIT/COMRESET and COMWAKE. Devices should send COMINIT in response to COMRESET. Devices that support asynchronous signal recovery may send out unsolicited COMINIT without receiving COMRESET and do not require an AWG stimulus. Hosts should always send out COMRESET at least once upon power up or reset. Hosts that support asynchronous signal recovery may also send out unsolicited COMRESET or in response to COMINIT. These acquisitions will also be used for OOB-03, OOB-04 and OOB-05 as appropriate.

Test Setup:

If using a PeRT3, set "Use PeRT3" to "Yes" in the setup tab after selecting a configuration. QualiPHY will prompt for the required connections when appropriate and automate the OOB tests using the PeRT3. See Appendix B for more details.

If using an AWG, preload the AWG with waveform files to generate the nominal COMINIT/COMRESET and nominal COMWAKE, see Appendix C for details. Use SMA cables to connect the AWG to the PUT's receive pair on the fixture. For the Teledyne LeCroy TF-SATA-C fixture use B+ & B- for devices or A+ & A- for hosts. Connect the other pair from the test fixture to C1 and C2 on the oscilloscope. Connect the AWG marker/trigger output to "Aux In" on the oscilloscope.

For hosts, this test should be done on the worst case port identified. See Reference [4] for details.

This test is only run once at the maximum interface rate of the product (1.5Gb/s or 3.0Gb/s).

Test Procedure:

1) Open the QualiPHY SATA PHY, TSG, OOB, RSG test suite. See Appendix B for details.
2) Select an appropriate configuration for the maximum supported speed with OOB-02 selected.
3) Run the test. If using a PeRT3 the test is executed automatically. If using an AWG continue below:
4) When prompted to acquire COMINIT/COMRESET connect the AWG to the PUT and the oscilloscope as described above in Test Setup. Then set the product to output COMINIT/COMRESET. For a device that does not support asynchronous signal recovery (or if unknown) load the COMRESET (nominal) waveform into the AWG set the output to be 500mVppd. For a device that does support asynchronous signal recovery no stimulus is required. For a host, reset the power to the device.
5) If necessary, adjust the horizontal offset on the oscilloscope so that the acquisition is centered on COMINIT. Press Single Trigger until a satisfactory waveform is acquired.

6) Return to QualiPHY and select “OK” to dismiss the message box.
7) When prompted to acquire COMWAKE repeat the steps above using the COMWAKE waveform. This time centering the acquisition on COMWAKE.
8) When all tests have completed, generate a report. Observe the results for OOB-02 in the report.

Observable Results:

The mean UI During OOB Signaling value shall be between 646.67 and 686.67ps.

Possible Problems:

Test OOB-03 – COMINIT/RESET and COMWAKE Transmit Burst Length

Purpose: To verify that the COMINIT/RESET and COMWAKE Transmit Burst Length of the PUT's transmitter is within the conformance limits.

References:

[1] SATA Standard, 7.4.2, Table 59 – OOB Specifications
[2] Ibid, 7.4.3.7.4 – COMINIT/COMRESET and COMWAKE Transmit Burst Length
[3] Ibid, 7.6.27 – OOB Signaling Tests
[4] SATA Interoperability Program Unified Test Document, 3.19.4

Resource Requirements:

Same as for OOB-01.

See Appendix A for details.

Last Modification: April 29, 2011

Discussion:

Reference [1] specifies the Transmitted Signal conformance limits for SATA products. This specification includes conformance limits for the COMINIT/RESET and COMWAKE Transmit Burst Length. Reference [2] provides the definition of this term for the purposes of SATA testing and specifies that the burst length should be measured in ns from the first crossing point of the burst at +/- 100 mV to the last crossing point of the burst at +/- 100 mV. Reference [3] defines the measurement requirements for this test.

The data already acquired, described in OOB-02, is used for this test. There is no setup change and no acquisition specifically for this test.

This test is only run once at the maximum interface rate of the product (1.5Gb/s, 3.0Gb/s or 6.0Gb/s).

Test Setup:

Same as for OOB-02.

Test Procedure:

1) Open the QualiPHY SATA PHY, TSG, OOB, RSG test suite. See Appendix B for details.
2) Select an appropriate configuration for the maximum supported speed with OOB-03 selected.
3) Run the test. If using a PeRT3 the test is executed automatically. If using an AWG continue below:
4) When prompted to acquire COMINIT. Load the COMINIT or COMWAKE (nominal) waveforms into the AWG set the output to be 500mVppd.
5) Connect the AWG to the PUT and the oscilloscope as described above in Test Setup.
6) Adjust the horizontal offset on the oscilloscope so that the acquisition is centered on COMINIT. The oscilloscope will stop acquiring when it recognizes COMINIT. Press Single Trigger until a satisfactory waveform is acquired.
7) Return to QualiPHY and select "OK" to dismiss the message box.
8) When prompted to acquire COMWAKE repeat the steps above this time centering the acquisition on COMWAKE.
9) When all tests have completed, generate a report. Observe the results for OOB-03 in the report.

Observable Results:

The COMINIT/RESET and COMWAKE Transmit Burst Length value shall be between 103.5ns and 109.9ns.

Possible Problems:

Test OOB-04 – COMINIT/RESET Transmit Gap Length

Purpose: To verify that the COMINIT/RESET Transmit Gap Length of the PUT's transmitter is within the conformance limits.

References:

[1] SATA Standard, 7.4.2, Table 59 – OOB Specifications
[2] Ibid, 7.4.3.7.5 – COMINIT/COMRESET Transmit Gap Length
[3] Ibid, 7.6.27 – OOB Signaling Tests
[4] SATA Interoperability Program Unified Test Document, 3.19.5

Resource Requirements:

Same as for OOB-02.

Last Modification: April 29, 2011

Discussion:

Reference [1] specifies the Transmitted Signal conformance limits for SATA products. This specification includes conformance limits for the COMINIT/RESET Transmit Gap Length. Reference [2] provides the definition of this term for the purposes of SATA testing and specifies that the gap length should be measured in ns from the last crossing point of the burst preceding the gap at +/- 100 mV to the first crossing point of the burst following the gap at +/- 100 mV. Reference [3] defines the measurement requirements for this test.

This test is only run once at the maximum interface rate of the product (1.5Gb/s, 3.0Gb/s or 6.0Gb/s).

Test Setup:

Same as for OOB-01.

Test Procedure:

1) Open the QualiPHY SATA PHY, TSG, OOB, RSG test suite. See Appendix B for details.
2) Select an appropriate configuration for the maximum supported speed with OOB-04 selected.
3) Run the test. If using a PeRT3 the test is executed automatically. If using an AWG continue below:
4) When prompted to acquire COMINIT. Load the COMINIT or COMWAKE (nominal) waveforms into the AWG set the output to be 500mVppd.
5) Connect the AWG to the PUT and the oscilloscope as described above in Test Setup.
6) Adjust the horizontal offset on the oscilloscope so that the acquisition is centered on COMINIT. The oscilloscope will stop acquiring when it recognizes COMINIT. Press Single Trigger until a satisfactory waveform is acquired.
7) Return to QualiPHY and select "OK" to dismiss the message box.
8) When prompted to acquire COMWAKE repeat the steps above this time centering the acquisition on COMWAKE.
9) When all tests have completed, generate a report. Observe the results for OOB-04 in the report.

Observable Results:

The COMINIT/RESET Transmit Gap Length value shall be between 310.4 ns and 329.6 ns.

Possible Problems:

Test OOB-05 – COMWAKE Transmit Gap Length

Purpose: To verify that the COMWAKE Transmit Gap Length of the PUT's transmitter is within the conformance limits.

References:

[1] SATA Standard, 7.4.2, Table 59 – OOB Specifications
[2] Ibid, 7.4.3.7.6 – COMWAKE Transmit Gap Length
[3] Ibid, 7.6.27 – OOB Signaling Tests
[4] SATA Interoperability Program Unified Test Document, 3.19.6

Resource Requirements:

Same as for OOB-01.

Last Modification: April 29, 2011

Discussion:

Reference [1] specifies the Transmitted Signal conformance limits for SATA products. This specification includes conformance limits for the COMWAKE Transmit Gap Length. Reference [2] provides the definition of this term for the purposes of SATA testing and specifies that the gap length should be measured in ns from the last crossing point of the burst preceding the gap at +/- 100 mV to the first crossing point of the burst following the gap at +/- 100 mV. Reference [3] defines the measurement requirements for this test.

This test is only run once at the maximum interface rate of the product (1.5Gb/s, 3.0Gb/s or 6.0Gb/s).

Test Setup:

Same as for OOB-02.

Test Procedure:

1) Open the QualiPHY SATA PHY, TSG, OOB, RSG test suite. See Appendix B for details.
2) Select an appropriate configuration for the maximum supported speed with OOB-05 selected.
3) Run the test. If using a PeRT3 the test is executed automatically. If using an AWG continue below:
4) When prompted to acquire COMINIT. Load the COMINIT or COMWAKE (nominal) waveforms into the AWG set the output to be 500mVppd.
5) Connect the AWG to the PUT and the oscilloscope as described above in Test Setup.
6) Adjust the horizontal offset on the oscilloscope so that the acquisition is centered on COMINIT. The oscilloscope will stop acquiring when it recognizes COMINIT. Press Single Trigger until a satisfactory waveform is acquired.
7) Return to QualiPHY and select "OK" to dismiss the message box.
8) When prompted to acquire COMWAKE repeat the steps above this time centering the acquisition on COMWAKE.
9) When all tests have completed, generate a report. Observe the results for OOB-05 in the report.

Observable Results:

The COMWAKE Transmit Gap Length value shall be between 103.5 ns and 109.9 ns.

Possible Problems:

Test OOB-06 – COMWAKE Gap Detection Windows

Purpose: To verify that the COMWAKE Gap Detection Windows of the PUT's receiver are within the conformance limits.

References:

[1] SATA Standard, 7.4.2, Table 52 – OOB Specifications
[2] Ibid, 7.4.3.7.7 – COMWAKE Gap Detection Windows
[3] Ibid, 7.6.27 – OOB Signaling Tests
[4] SATA PRE-TEST MOI
[5] SATA Interoperability Program Unified Test Document, 3.19.7

Resource Requirements:

Same as OOB-01.

See Appendix A for details.

Last Modification: April 29, 2011

Discussion:

Reference [1] specifies the Transmitted Signal conformance limits for SATA products. This specification includes conformance limits for the COMWAKE Gap Detection Windows. Reference [2] provides the definition of this term for the purposes of SATA testing. Reference [3] defines the measurement requirements for this test.

This test is run by sending the PUT a nominal COMINIT followed by a COMWAKE in which the OOB gap timing is not nominal and observing if the PUT responds. Four different OOB gap timings are used: $45UI_{OOB}$ (30ns), $153UI_{OOB}$ (102ns), $167UI_{OOB}$ (111.3ns) and $266UI_{OOB}$ (177ns). When the PUT detects COMWAKE it will bring up the link and continuous data will be transmitted.

This test is only run once at the maximum interface rate of the product (1.5Gb/s, 3.0Gb/s or 6.0Gb/s).

Test Setup:

If using a PeRT3, set "Use PeRT3" to "Yes" in the setup tab after selecting a configuration. QualiPHY will prompt for the required connections when appropriate and automate the OOB tests using the PeRT3. See Appendix B for more details.

If using an AWG, preload the AWG with waveform files to generate COMWAKE for each of the 4 gap timings, see Appendix C for details. Use SMA cables to connect the AWG to the PUT's receive pair on the fixture. For the ICT fixture use 2 & 3 for devices or 5 & 6 for hosts. Connect the other pair from the test fixture to C1 and C2 on the oscilloscope. Connect the AWG marker/trigger output to C4 on the oscilloscope.

Test Procedure:

1) Open the QualiPHY SATA PHY, TSG, OOB, RSG test suite. See Appendix B for details.
2) Select an appropriate configuration for the maximum supported speed with OOB-06 selected.
3) Run the test. If using a PeRT3 the test is executed automatically. If using an AWG continue below:
4) When prompted to run Test OOB-06 with the $45UI_{OOB}$ (30ns) gap timing, load the COMWAKE waveform with $45UI_{OOB}$ (30ns) gap timing into the AWG. If the PUT responds to COMINIT, but not to COMWAKE the COMINIT will be visible as very short bursts on C1 and C2 at the same interval as the marker on C4, as shown in Figure 6. If the PUT detects COMWAKE large bursts of data will be seen after each marker as in Figure 7. If the PUT detects COMWAKE select "Detect" in the message box. Otherwise, select "No Detect."
5) Load the waveforms for $153UI_{OOB}$ (102ns), $167UI_{OOB}$ (110.3ns) and $266UI_{OOB}$ (177ns) when prompted and select "Detect" or "No Detect" as in Step 4.

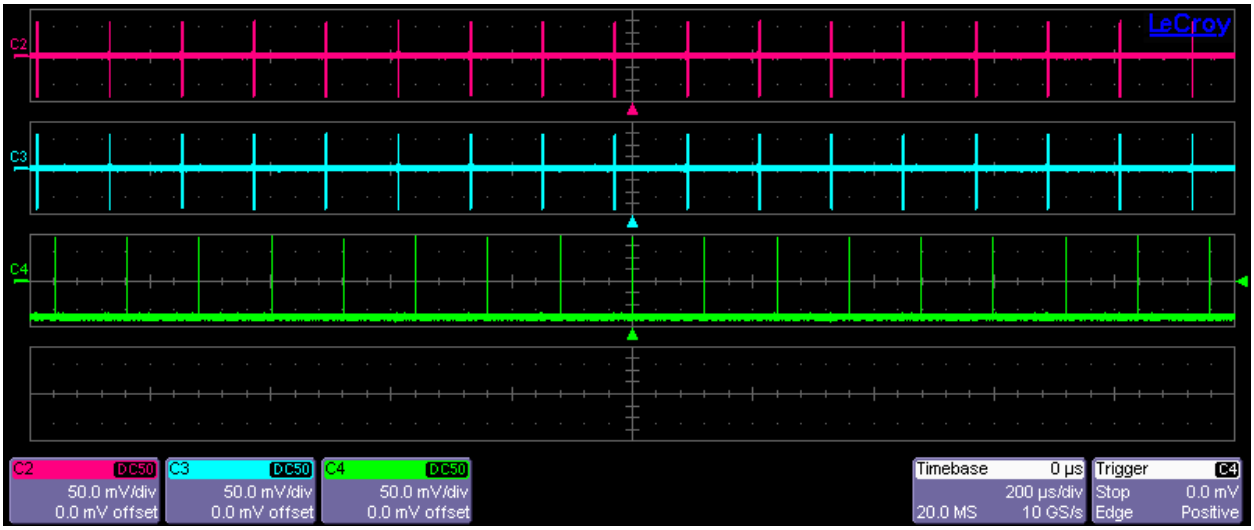


Figure 6. Image of Test OOB-06 showing a PUT that does not detect COMWAKE

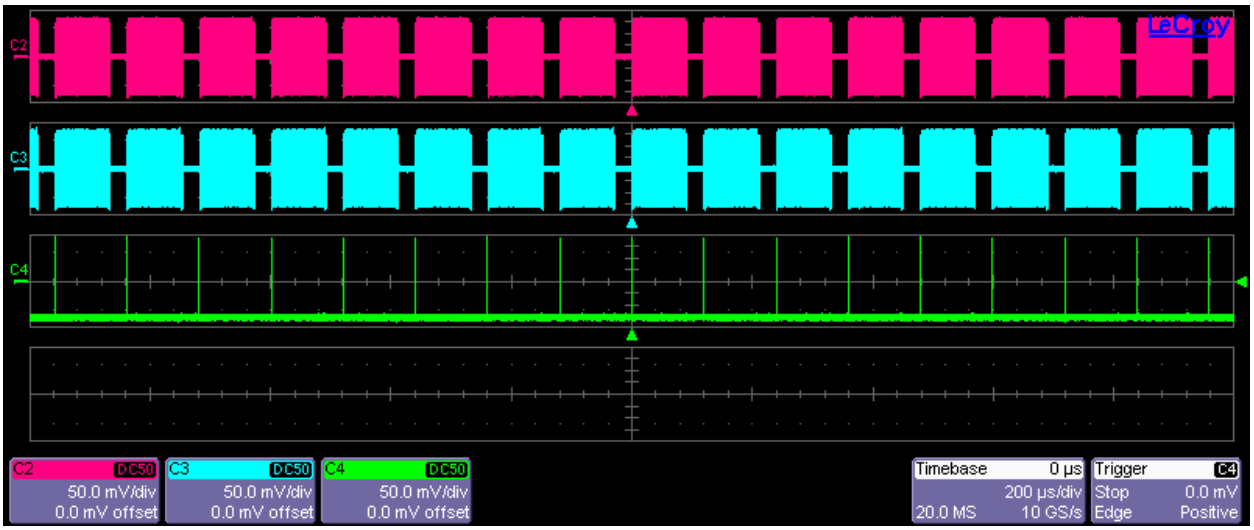


Figure 7. Image of Test OOB-06 showing a PUT that detects COMWAKE

Observable Results:

The PUT shall detect COMWAKE at the lower limit of $153UI_{OOB}$ (102ns).
The PUT shall detect COMWAKE at the upper limit of $167UI_{OOB}$ (111.3ns).
The PUT shall **not detect** COMWAKE at the lower limit of $45UI_{OOB}$ (30ns).
The PUT shall **not detect** COMWAKE at the lower limit of $266UI_{OOB}$ (177ns).

Possible Problems:

Test OOB-07 – COMINIT Gap Detection Windows

Purpose: To verify that the COMINIT Gap Detection Windows of the PUT's receiver are within the conformance limits.

References:

[1] SATA Standard, 7.4.2, Table 59 – OOB Specifications
[2] Ibid, 7.4.3.7.8 – COMINIT/COMRESET Gap Detection Windows
[3] Ibid, 7.6.27 – OOB Signaling Tests
[4] SATA Interoperability Program Unified Test Document, 3.19.8

Resource Requirements:

Same as OOB-01.

See Appendix A for details.

Last Modification: April 29, 2011

Discussion:

Reference [1] specifies the Transmitted Signal conformance limits for SATA products. This specification includes conformance limits for the COMINIT Gap Detection Windows. Reference [2] provides the definition of this term for the purposes of SATA testing. Reference [3] defines the measurement requirements for this test.

This test is run by sending the PUT a COMINIT in which the OOB gap timing is not nominal and observing if the PUT responds. Four different OOB gap timings are used: $259UI_{OOB}$ (173ns), $459UI_{OOB}$ (306ns), $501UI_{OOB}$ (334ns) and $791UI_{OOB}$ (527ns). When a device detects COMINIT it should respond with COMINIT. When a host detects COMINIT it should respond with COMWAKE.

This test is only run once at the maximum interface rate of the product (1.5Gb/s, 3.0Gb/s or 6.0Gb/s).

Test Setup:

If using a PeRT3, set "Use PeRT3" to "Yes" in the setup tab after selecting a configuration. QualiPHY will prompt for the required connections when appropriate and automate the OOB tests using the PeRT3. See Appendix B for more details.

If using an AWG, preload the AWG with waveform files to generate COMINIT for each of the 4 gap timings, see Appendix C for details. Use SMA cables to connect the AWG to the PUT's receive pair on the fixture. For the Teledyne LeCroy TF-SATA-C fixture use B+ & B- for devices or A+ & A- for hosts. Connect the other pair from the test fixture to C1 and C2 on the oscilloscope. Connect the AWG marker/trigger output to C4 on the oscilloscope.

Test Procedure:

1) Open the QualiPHY SATA PHY, TSG, OOB, RSG test suite. See Appendix B for details.
2) Select an appropriate configuration with OOB-07 selected.
3) Run the test. If using a PeRT3 the test is executed automatically. If using an AWG continue below:
4) When prompted to run Test OOB-07 with the $259UI_{OOB}$ (173ns) gap timing, load the COMINIT waveform with $259UI_{OOB}$ gap timing into the AWG. Verify that for every COMINIT sent by the AWG there is a response from the PUT. There should be a response to COMINIT from the PUT following every marker output as shown above in Figure 5 for Test OOB-01. If the product always responds to COMINIT select "Detect" in the message box. Otherwise, select "No Detect."
5) Load the waveforms for $459UI_{OOB}$ (306ns), $501UI_{OOB}$ (334ns)and $791UI_{OOB}$ (527ns) when prompted and select "Detect" or "No Detect" as in Step 4.

Observable Results:

The PUT shall detect COMINIT/RESET at the lower limit of $459UI_{OOB}$ (306ns).
The PUT shall detect COMINIT/RESET at the upper limit of $501UI_{OOB}$ (334ns).

The PUT shall **not detect** COMINIT/RESET at the lower limit of $259UI_{OOB}$ (173ns).
The PUT shall **not detect** COMINIT/RESET at the lower limit of $791UI_{OOB}$ (527ns).

Possible Problems:

Appendix A – Information on Required Resources

Equipment referred to in this document is described here, or references to available resources are cited.

The following table lists the oscilloscopes that are suitable for SATA compliance testing:

Oscilloscope Model	Supported Interface Tests
SDA6000, SDA9000, SDA760Zi-A, SDA806Zi-A, SDA808Zi-A	Gen1 only
SDA11000	Gen1 and Gen2 only
SDA13000, SDA813Zi-A, SDA816Zi-A or higher bandwidth	Gen1, Gen2 and Gen3

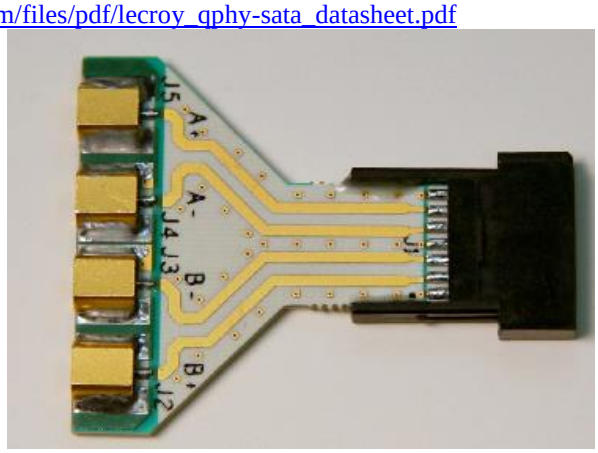
The SDA-8Zi-A Series specifications and description can be found in its brochure, on the Teledyne LeCroy web site at: http://www.teledynelecroy.com/files/pdf/lecroy_wavemaster_8_zi-a_datasheet.pdf

Selected oscilloscope characteristics that effect accuracy are provided below for convenience (for SDA813Zi-A):

Oscilloscope Characteristic	Value
Analog Bandwidth @ 50Ω (-3dB)	**13GHz**
DC Gain Accuracy	**± 1.5 % of full scale**
Sample Rate and Delay Time Accuracy	**± 1 ppm ≤ 10 sec interval (typical)**
Jitter Noise Floor	**< 315 fs rms**

Suitable SMA cables are two 30” SEMFLEX SW180 SMA cable assemblies (21 21 SW180 030). These are double shielded low loss cables for use up to 18GHz. More information on these cable assemblies is available from SEMFLEX’s web site, at:
http://www.semflex.com/pdf/SWI180.pdf
Note that the cables are not a precision matched pair. Therefore setting a Deskew value on the SDA-11000 to deskew the cables is required. A procedure for doing this is supplied below in this appendix, after the description of the other resources listed in this document. Higher precision and/or higher bandwidth cables may be used, of course.

Test Fixtures
There are several acceptable SATA test fixtures that may be used. The Teledyne LeCroy TF-SATA-C Test Fixture is shown below. The host TX pair is A+ and A-, and the device TX pair is B+ and B-. For more information see the QPHY-SATA data sheet on the Teledyne LeCroy website at:
http://www.teledynelecroy.com/files/pdf/lecroy_qphy-sata_datasheet.pdf



Teledyne LeCroy TF-SATA-C Test Fixture

ICT Solutions also offers a SATA Test fixture that may be used. A close up picture of the ICT Solutions TF-1R31 SATA Receptacle Gen 3 Test Fixture is below, note that the SMA connectors are labeled 2, 3, 5 and 6. The host TX pair is 2 & 3, and the device TX pair is 5 & 6. For more information or to order an ICT fixture visit their website at:
http://www.ictsols.com/



Wilder Technologies also offers a SATA Test fixture that may be used. A close up picture of the Wilder Techologies SATA Gen3 Receptacle is shown below. The host TX pair is Drive RP+ and Drive RP- and the device TX pair is Drive TP+ and Drive TP-. For more information or to order a Wilder Techologies fixture visit their website at:
http://www.wilder-tech.com/



NOTE: The SATA cable end connector on the fixture is fragile! Support the cables connected to the fixture. Do not let their weight be supported by torque on the SATA connector. When unplugging the fixture grasp the sides of the SATA connector, not the cables or SMA connectors!

BIST Initiator Tools
Teledyne LeCroy offers protocol analyzers as well as a Protocol-enabled Receiver and Transmitter Tolerance Tester (PeRT^3) that may be used to place a device or host into the required BIST modes.

Teledyne LeCroy recommends the PeRT^3 for use as a BIST initiator tool. It also supports the running the OOB tests and Receive Signal (RSG) tests thus reducing costs of an additional signal generator. The Teledyne LeCroy PeRT3 supports SATA up to 6Gb/s. It can be used for other protocols as well including, SAS, USB3 and PCI-Express. Information about the PeRT^3 can be found on the Teledyne LeCroy website at:
http://www.teledynelecroy.com/pert3/
Teledyne LeCroy also offers a large selection of protocol analyzer tools that may be used as BIST initiator tools. For testing products up to 6Gb/s Teledyne LeCroy offers the Sierra M6-2 or M6-4

Protocol Test System. For testing products that support a maximum speed of 3Gb/s or less the Teledyne LeCroy SATracer or SASTracer may be used. Information about Teledyne LeCroy's complete line of SATA protocol analyzer and exerciser tools can be found on the Teledyne LeCroy website at: http://www.teledynelecroy.com/ProtocolAnalyzer/ProtocolStandard.aspx?standardID=8&capid=103&mid=511

The SATA QualiPHY test suite includes generation scripts for the SASTracer/Trainer to place a host or device into BIST-T mode.

Cable Deskew Procedure

This procedure must be performed before measurements are made, and whenever the deskew requirement may have changed (i.e., cables have been disconnected and reconnected perhaps on the other side of the diff pair). We recommend that the oscilloscope should be thermally stable, i.e. turned on for at least 15 minutes, before performing this procedure.

Important note: SMA connectors should always be tightened and removed with a calibrated torque wrench for that purpose. The torque wrench should limit torque to <= 1Nm.

This procedure is easiest to perform when a SATA product is producing a repetitive signal such as HFTP. It is not pattern dependent; however, any pattern is usable. All that is needed is a signal. When running QualiPHY, after using the Tracer/Trainer to make the PUT produce HFTP, when prompted to reconnect the oscilloscope, it is possible to do the deskew procedure at that time before pressing "OK". The deskew procedure is:

1) Insert the SATA test fixture into the product producing the pattern.
2) Connect the SATA test fixture to the oscilloscope using the two supplied SMA cables. Connect C1 to 6 and C2 to 5.
3) On the menu bar, select Measure then Measure Setup. On the Measure menu, make the Statistics On checkbox checked, to turn on the Statistics.
4) On the Measure menu select the tab of an unused parameter; in this procedure we will assume P8.
5) Set P8 to be Skew of C1 to C2. On the right side dialog for Skew, set Slope to Both for both inputs.
6) Press AUTO trigger on the front panel; set C1 and C2 to 100mV/div, offset appropriate to center the signal vertically (should be 0V for a product running at 3.0Gb/s) and set the timebase to 10ns/div or so.
7) Allow the statistics to accumulate for a few seconds. Write down the "mean" skew value.
8) Disconnect the cables from the test fixture and swap the ends. Now the cable that had been connected from C2 to 6 should be connected from C1 to 5 and C2 should be connected to 6. Then press Clear Sweeps on the SDA-11000's front panel.
9) Allow the statistics to accumulate for a few seconds. Write down the "mean" skew value.
   The first measured skew value is: device skew + cable skew.
   The second measured skew value is: device skew - cable skew.
   Subtract the second mean value from the first mean value. The result is twice the cable skew. Divide the result by two to obtain the deskew value.
10) Skew measures time of source1 (C1) edge minus time of source2 (C2) edge. Therefore a positive value means that the delay through the cable to C1 is less than the delay through the cable to C2. Therefore, to cancel the cable skew, set the value found in step 10 into the "Deskew" setting on C2's menu.

Appendix B – Using the Teledyne LeCroy QualiPHY SATA Test Suite

Teledyne LeCroy QualiPHY is a collection of test suites that are run by the X-Replay test engine. QualiPHY may be installed on any PC or directly on the oscilloscope. To launch QualiPHY double-click the QualiPHY icon on the desktop or the Start Menu. On a oscilloscope, it can also be launched from the Analysis menu. To run the tests described in this document select "SATA TSG, RSG" as the standard.

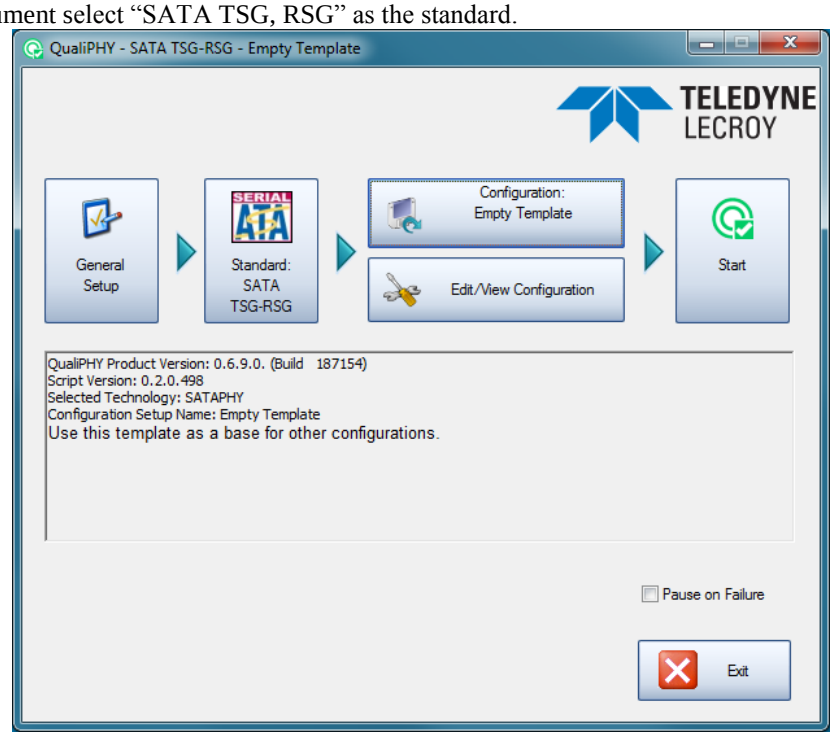


Figure B-1 QualiPHY Main Window

The SATA TSG, RSG QualiPHY script is preloaded with configurations for common test setups. These locked configurations are supplied by Teledyne LeCroy and cannot be modified, so they will always be available. A configuration contains information about what tests to run, what limit criteria to use and other details specific to the test suite. The configuration details that a user can modify for the SATA PHY, TSG, OOB, RSG test suite include such things as whether SSC is enabled, the directory to which acquired waveforms are stored, whether the test will be conducted on acquired data or on previously stored data, etc. These include devices running at 1.5 Gb/s, 3.0 Gb/s and 6.0Gb/s. Select the configuration that most closely matches the product under test. Click "Edit/View Configuration". A new dialog appears which allows you to modify the configuration as needed. The "Setup" dialog is shown in Figure B-2 below.

In this dialog, set the Product Type to device or host. Set the speed to be tested and the product SSC setting. Modify any other settings as necessary. If a PeRT3 will be used for testing ensure that "Use PeRT3" is set to yes. Check the groups of tests to be tested. To enable or disable individual tests click on the "Test Selector" tab and select the tests you wish to run, see Figure B-3.

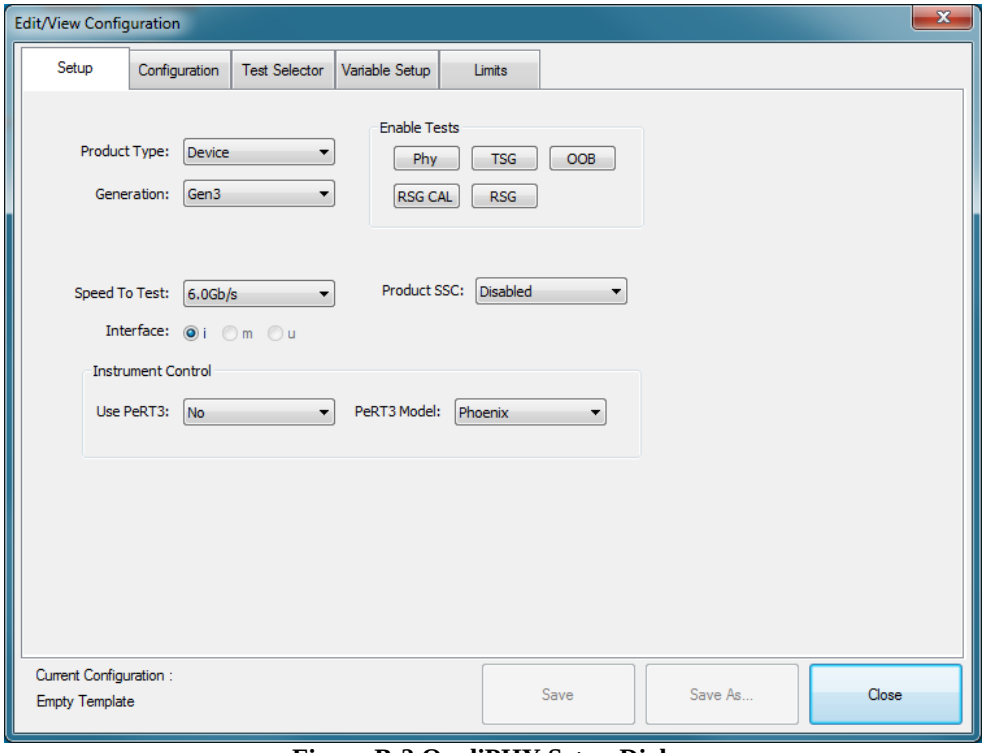


Figure B-2 QualiPHY Setup Dialog

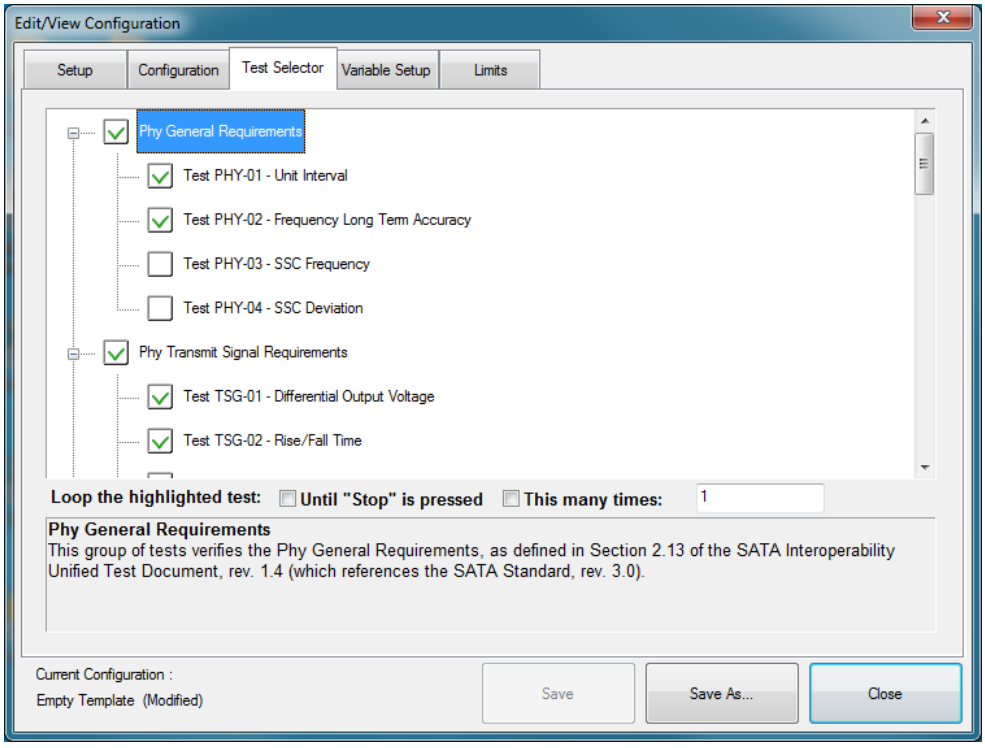


Figure B-3 QualiPHY Test Selection

The “Variable Setup” tab (Figure B-4) allows the user to change more advanced details of the selected configuration. The variables that can be modified are shown next to “gear” icons. To change a variable of the setup

simply double click it or click the “Edit Variable” button. A short description of the selected variable is provided in the text box at the bottom of the dialog. Set the “SSC Setting” to the appropriate value for the product under test.

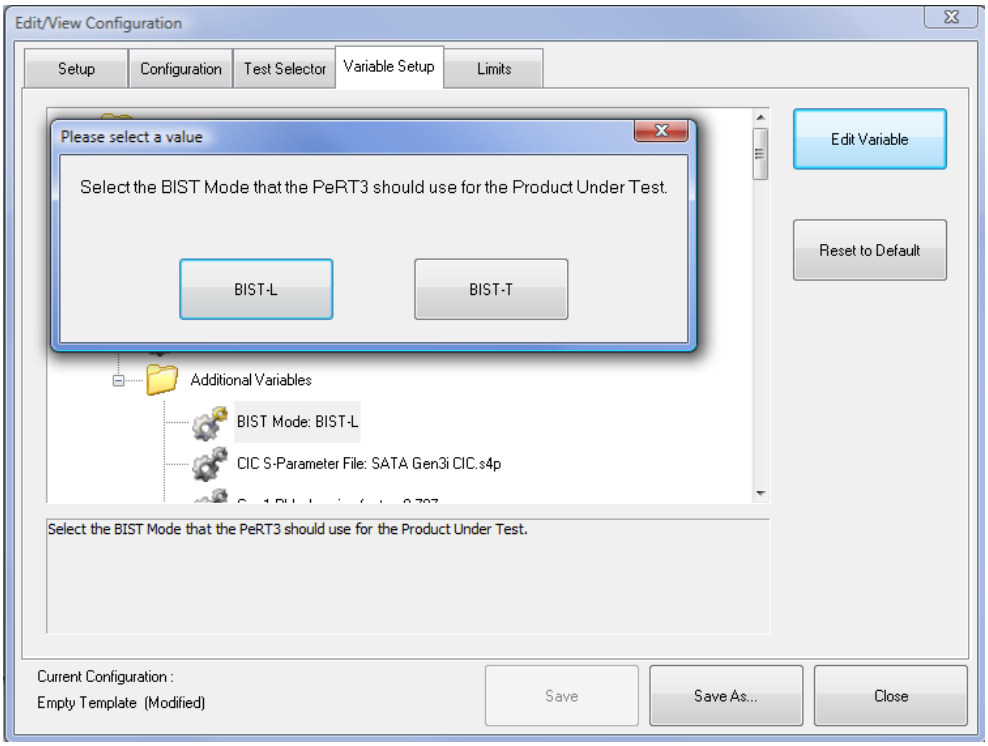


Figure B-4 QualiPHY Variable Setup View

Limits can modified from the “Limits” tab. The provided limit sets cannot be modified, so that the official sets are always available. A user can create a custom limit set by copying an existing limit set and editing it. When finished editing the configuration return to the “Configuration” tab and click “Stop Edit”. Then click “Close” to close the “Edit/View Configuration”.

When all settings are as desired the configuration can be saved for future use by clicking the “Save As…” button at the bottom.

Once either a predefined configuration or a customized configuration has been selected click “Start” to begin testing. When testing is complete use the report generator to create a report containing all the test results. For further details about using QualiPHY consult the QualiPHY Operator’s Manual.

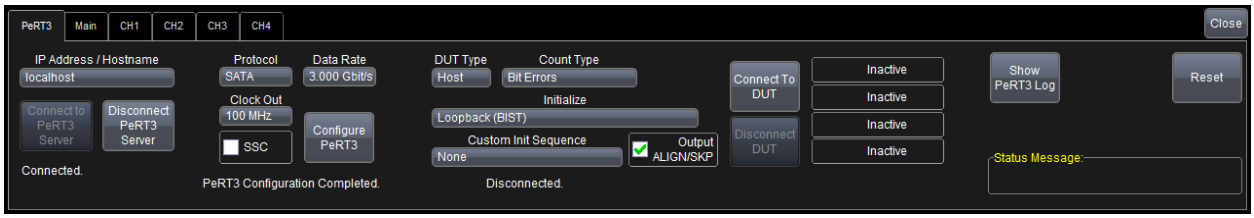
Appendix C – Procedures for Manual Operation

The QualiPHY SATA test suite requires the use of additional instruments that are not automated by QualiPHY. Several instruments may be used as a BIST Initiator including a Teledyne LeCroy PeRT^3, Sierra Protocol Test System or SAS Tracer/Trainer. Other tools may be used as well but are not documented here. This section describes the test procedures for manual operation of the Teledyne LeCroy BIST tools only.

Using the PeRT^3 to Place the PUT into BIST-L Mode

The PeRT^3 can easily place a device or host into a BIST-L (loopback) mode. First the PeRT^3 software must be installed on the oscilloscope and the PeRT^3 unit should be connected to the oscilloscope via USB cable. Connect the PeRT^3 TX and RX connections to the SATA Test Fixture and plug the fixture into the PUT. For some devices or hosts a Teledyne LeCroy SATA Switch may be required in between the PeRT^3 TX outputs and the SATA Test Fixture. Once the connections are made the PeRT^3 can be operated from the oscilloscope UI as follows:

1) Click **Analysis -> PeRT^3** to bring up the PeRT^3 control dialog.
2) Set the **Hostname** to "localhost" and click "**Connect to PeRT^3 Server**"
3) Select "SATA" for **Protocol**. Set the desired Data Rate (1.5Gb/s, 3.0Gb/s or 6.0Gb/s). Set the desired SSC setting for the PeRT^3 and click **Configure PeRT^3**.
4) Set **DUT Type** to either "Device" or "Host." Set **Initialize** to "Loopback (BIST)." The dialog should look similar to this:



5) Click **Connect to DUT.** This should place the PUT into BIST-L loopback, and the dialog should indicate "Pattern Locked."
6) Click on the CH1 tab. Adjust the **Amplitude** (0.4V should be sufficient) and select the desired pattern.
7) Click on the Main tab. Set **Termination Type** to "Run Forever."
8) Click **Start** to begin transmitting the selected pattern.
9) Disconnect the SMA cables from the PeRT^3 RX connectors and attach them to the oscilloscope to view the output of the PUT. Moving the SMA connections one at a time should prevent the PUT from dropping out of loopback.

To change the pattern at any time return to the CH1 tab of the PeRT^3 dialog and select a new pattern.

Using the SASTracer to Place the PUT into BIST-T Mode

It is possible to control the SAS Tracer/Trainer from its software interface either on the oscilloscope or on a PC. This section uses SASTracer script files which are installed as part of the QualiPHY SATA test suite. By default these files are installed under:

"C:\Program Files\LeCroy\XReplay\SATA\SASTracer"

The QualiPHY SATA test suite prompts the user to produce the various required patterns. In this example HFTP will be produced; the other Trainer generation scripts are similarly named. To generate a pattern manually, do the following:

1) For a HDD: For Gen1 (1.5Gb/s) testing, load the Trainer script LeCroy BIST HFTP.ssg from the "BIST Scripts" folder. For Gen2 (3.0Gb/s) testing, load the Trainer script LeCroy 3G BIST HFTP.ssg. For an ATAPI drive: For Gen1 testing, load "LeCroy ATAPI BIST HFTP.ssg". For Host testing load "LeCroy Host BIST HFTP.ssg" or for Gen2 speed "LeCroy3G Host BIST HFTP.ssg".
2) To check proper operation, set Record options to Trigger on BIST Activate FIS. Start recording.
3) Start Generation. If the Tracer triggers then the script ran to completion, the Trainer transmitted the BIST Activate FIS. Otherwise, power off the product, power it on; and repeat this step.

4) OBSERVE the signal on the oscilloscope. If it is not HFTP, the product did not properly handle BIST Activate FIS; a non-standard way to make it produce the desired pattern will be required.

OOB Test Procedures Using the PeRT[3]

The procedure for running an OOB tests is nearly the same as the procedure for placing a product in BIST-L described above. To run an OOB test on a product follow the BIST-L procedure, but set **Initialize** to "Custom Sequence" and choose the desired OOB test in **Custom Init Sequence**. Click **Connect To DUT** to run the test (the **Disconnect DUT** button will remain grayed out). The result is reported in the Status Message. The "OOB mV calibration" sequence will repeatedly output a COMINIT/COMRESET to allow for amplitude calibration. The other sequences send out the patterns 1000 times. The product should report 1000 passes.

OOB Test Procedures using an Arbitrary Waveform Generator

An arbitrary waveform generator (AWG) can be used to generate the COMINIT and COMWAKE signals needed for Tests OOB-01 – OOB-07. Each of the required waveforms should be saved in the AWG before running the test so they will be available at run time. COMINIT and COMWAKE both consist of 6 bursts of 160 UIOOB each. A UIOOB is 1 UI at 1.5Gb/s, i.e. 1 UIOOB = 666.667ps. A burst can be created by repeating 4 ALIGN primitives. The ALIGN primitive is defined as: K28.5 D10.2 D10.2 D27.3. The 8b/10b encoding using positive running disparity from bit 0 to bit 39 is as follows:

0	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**12**	**13**	**14**	**15**	**16**	**17**	**18**	**19**
1	1	0	0	0	0	0	1	0	1	0	1	0	1	0	1	0	1	0	1

20	**21**	**22**	**23**	**24**	**25**	**26**	**27**	**28**	**29**	**30**	**31**	**32**	**33**	**34**	**35**	**36**	**37**	**38**	**39**
0	1	0	1	0	1	0	1	0	1	0	0	1	0	0	1	1	1	0	0

To create a single burst, repeat the above 40 bits 4 times.

To use an AWG for tests OOB-01 – OOB-07, create the following differential waveforms and save them to the AWG:

COMINITDevice (nominal) = 6 x (burst + 480 UI_{OOB} gap) + 1 x (45,000 UI_{OOB} gap)
COMINITHost (nominal) = 6 x (burst + 480 UI_{OOB} gap) + 1 x (300,000 UI_{OOB} gap)
COMINIT259 = 6 x (burst + 259 UI_{OOB} gap) + 1 x (45,000 UI_{OOB} gap)
COMINIT459 = 6 x (burst + 459 UI_{OOB} gap) + 1 x (45,000 UI_{OOB} gap)
COMINIT501 = 6 x (burst + 501 UI_{OOB} gap) + 1 x (45,000 UI_{OOB} gap)
COMINIT791 = 6 x (burst + 791 UI_{OOB} gap) + 1 x (45,000 UI_{OOB} gap)
For each of the above waveforms set the marker to produce pulse at the end of COMINIT to enable the oscilloscope to trigger at the end of the transmitted COMINIT.

COMWAKE (nominal) = COMINIT (nominal, host or device as appropriate) + 6 x (burst + 160 UI_{OOB} gap) + 1 x (130,000 UI_{OOB} gap)
COMWAKE45 = COMINIT (nominal) + 6 x (burst + 45 UI_{OOB} gap) + 1 x (130,000 UI_{OOB} gap)
COMWAKE153 = COMINIT (nominal) + 6 x (burst + 153 UI_{OOB} gap) + 1 x (130,000 UI_{OOB} gap)
COMWAKE167 = COMINIT (nominal) + 6 x (burst + 167 UI_{OOB} gap) + 1 x (130,000 UI_{OOB} gap)
COMWAKE266 = COMINIT (nominal) + 6 x (burst + 266 UI_{OOB} gap) + 1 x (130,000 UI_{OOB} gap)
For each of the above waveforms set the marker to produce pulse at the end of COMWAKE to enable the oscilloscope to trigger at the end of the transmitted COMWAKE.

Appendix D – Verification of Lab Load Requirements

The SATA specification indicates that products must be tested using a "Lab Load', which is defined to include the inputs of the test equipment as well as any cable connected. The test fixture is not included in the lab load. The requirement for the lab load is that each input must have a return loss of greater than 20 dB over a bandwidth of 100 MHz to 5 GHz and greater than 10 dB from 5 GHz to 8 GHz, see section 7.2.2.5 of the SATA Standard. In order to meet this specification external attenuators need to be attached to the SDA11000. A 6 dB attenuator is recommended. Any instrument capable of measuring return loss up to 8 GHz can be used to verify the return loss of the SDA11000. To use a Teledyne LeCroy WaveExpert with TDR to verify the return loss, follow these steps:

1) Click "**Timebase"** on the menu, and then click "**TDR**."
2) Click "**Calibrate**" to launch the TDR Calibration Wizard
3) Follow the wizard steps to complete a 1 port, single-ended calibration. The reference plane should be at the ends of the cables used for testing.
4) When the calibration is complete change the measurement to S11 in the TDR Normalization dialog, as shown below:

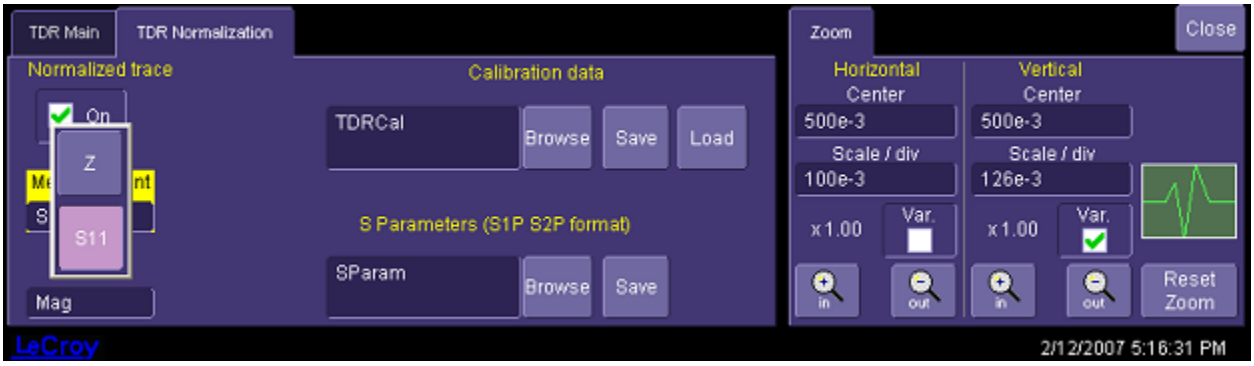


Figure D-1. TDR Normalization Dialog showing selection of S11 measurement

5) Measure the return loss of both channels. Connect attenuators to the oscilloscope channel inputs and verify that it meets the return loss requirements with the attenuators attached.
6) When running a test with attached attenuators make sure to set the "External Attenuation" variable in the QualiPHY configuration as described in Appendix B.

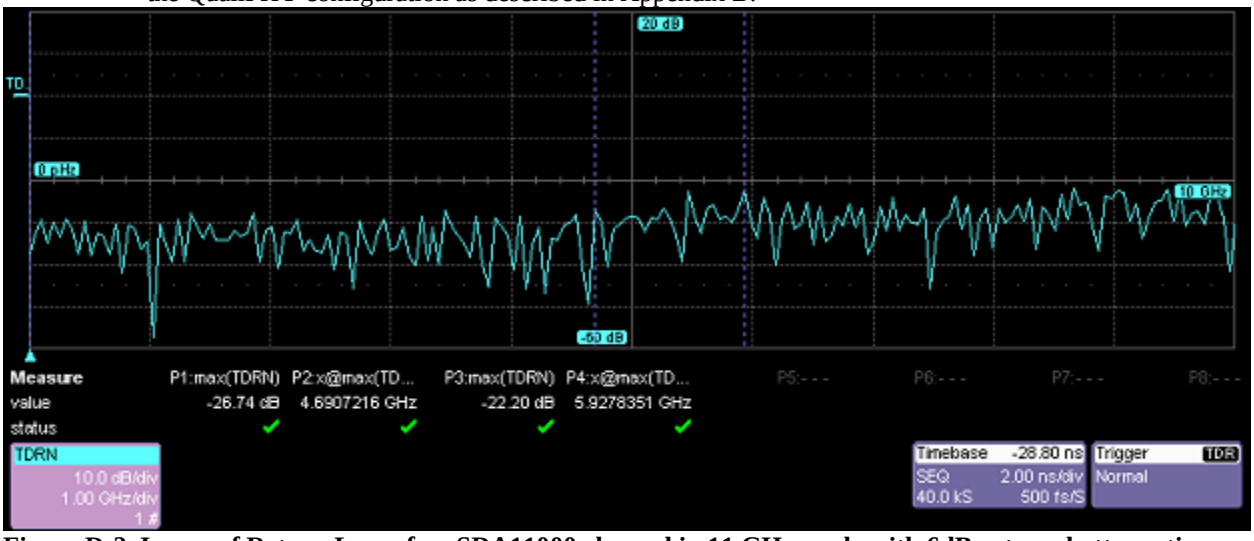


Figure D-2. Image of Return Loss of an SDA11000 channel in 11 GHz mode with 6dB external attenuation. P1 shows the max return loss below 5 GHz. P3 shows the max return loss between 5 and 8 GHz.

Appendix E – Calibration and Verification of Jitter Measurement Devices

Purpose: To calibrate and verify the jitter measurement device (JMD) and associated test setup has a proper response to jitter and SSC. Currently these JTF considerations are only for the following interfaces, Gen1i, Gen1m, Gen2i, Gen2m and Gen3i.

References:

[1] SATA Interoperability Program Unified Test Document, 6.6

Resource requirements:

Pattern Generator for SATA signals
Sine wave source, 30kHz, and 0.5MHz to 50MHz.
Test cables
Jitter Measuring Device

Last Modification: March 25, 2009

Discussion:

See Reference [1].

Test Procedure:

The following procedure is based on the test procedure in [1] but with specific instructions for Teledyne LeCroy oscilloscopes.

The response to jitter of the Jitter Measurement Device (JMD)(the reference clock is part of the JMD) is measured with three different jitter modulation frequencies corresponding to the three cases: 1) SSC (full tracking) 2) jitter (no tracking) 3) the boundary between SSC and jitter. The jitter source is independently verified by separate means. This ensures the jitter response of the JMD is reproducible across different test setups.

The three Gen1 test signals are: 1) a 375MHz +/- 0.035% square wave (which is a D24.3, 00110011 pattern) with risetime between 67ps and 136ps (20% to 80%) [1] with a sinusoidal phase modulation of 20.8ns +/- 10% peak to peak at 30kHz +/- 1%. 2) a 375MHz square wave with a sinusoidal phase modulation of 200ps +/- 10% peak to peak at 50MHz +/- 1%. 3) a 375MHz square wave with no modulation.

The three Gen2 test signals are: 1) a 750MHz +/- 0.035% square wave (which is a D24.3, 00110011 pattern) with risetime between 67ps and 136ps (20% to 80%) [1] with a sinusoidal phase modulation of 20.8ns +/- 10% peak to peak at 30kHz +/- 1%. 2) a 750MHz square wave with a sinusoidal phase modulation of 100ps +/- 10% peak to peak at 50MHz +/- 1%. 3) a 750MHz square wave with no modulation.

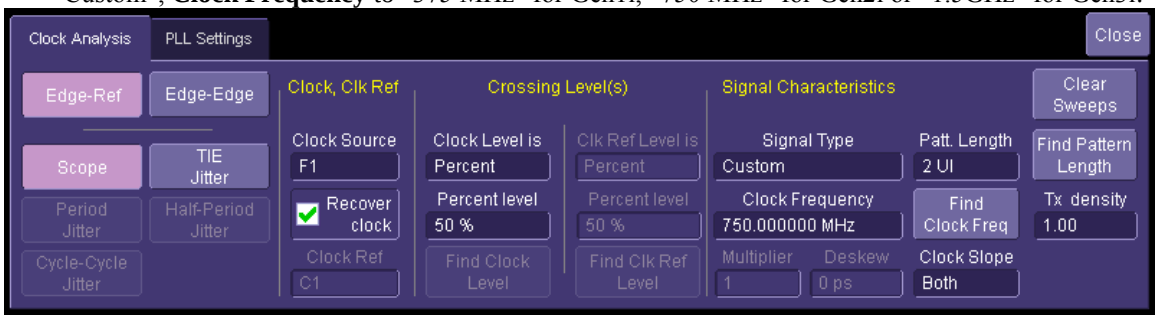
The three Gen3i test signals are: 1) a 1500MHz +/- 0.035% square wave (which is a D24.3, 00110011 pattern) with risetime between 33ps and 67ps (20 to 80%) [1] with a sinusoidal phase modulation of 1.0ns +/- 10% peak to peak at 420kHz +/- 1%. 2) a 1500MHz square wave with a sinusoidal phase modulation of 50ps +/- 10% peak to peak at 50MHz +/- 1%. 3) a 1500MHz square wave with no modulation.

An independent separate means of verification of the test signals is used to make sure the level of the modulation is correct.

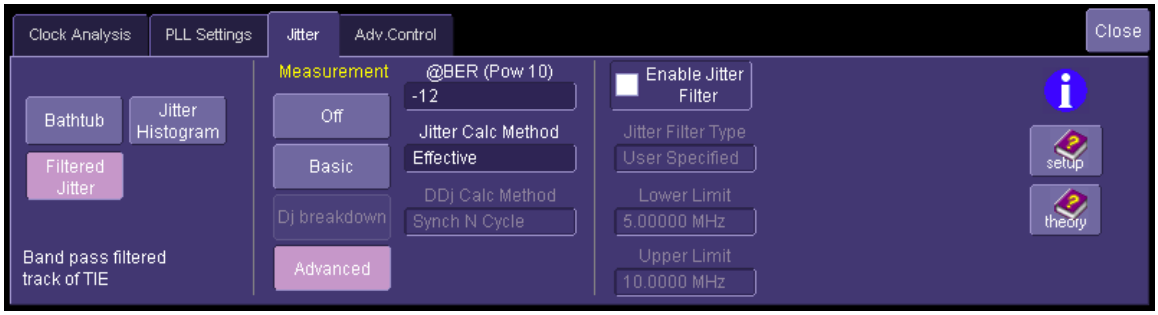
The test procedure checks two conditions: the JTF attenuation and the JTF bandwidth. Care is taken to minimize the number of absolute measurements taken, making most relative; this reduces the dependencies and improves accuracy.

1. Setup the oscilloscope as it will be used during testing: attach attenuators to C2 and C3, set probe attenuation to correct value, set C2 and C3 to 50 mV/div and deskew C2 and C3.
2. For Gen 1 and Gen 2 calibration, adjust the pattern generator for a D24.3 pattern (00110011, with a risetime within specified limits) modulation to produce a 30 KHz +/- 1%, 20.8 ns p-p +/- 10% sinusoidal phase modulation. For Gen3 calibration, adjust the pattern generator for a D24.3 pattern (00110011, with a risetime within specified limits) modulation to produce a 420kHz +/- 1%, 1.0 ns p-p +/- 10% sinusoidal phase modulation.

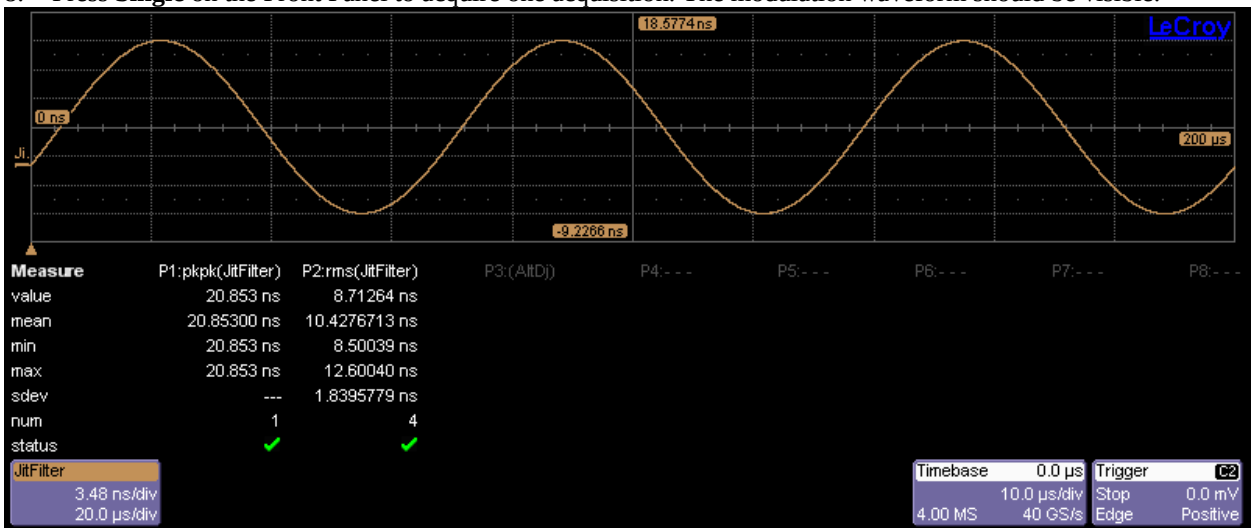
3. On the oscilloscope, set F1 as C2 – C3. In **Timebase** set to 11 GHz, 40 GS/s. Set Time/division to 10 us/div.
4. On the oscilloscope menu, click **Analysis -> Clock Analysis…** Set **Clock Source** to "F1", **Signal Type** to "Custom", **Clock Frequency** to "375 MHz" for Gen1i, "750 MHz" for Gen2i or "1.5GHz" for Gen3i.



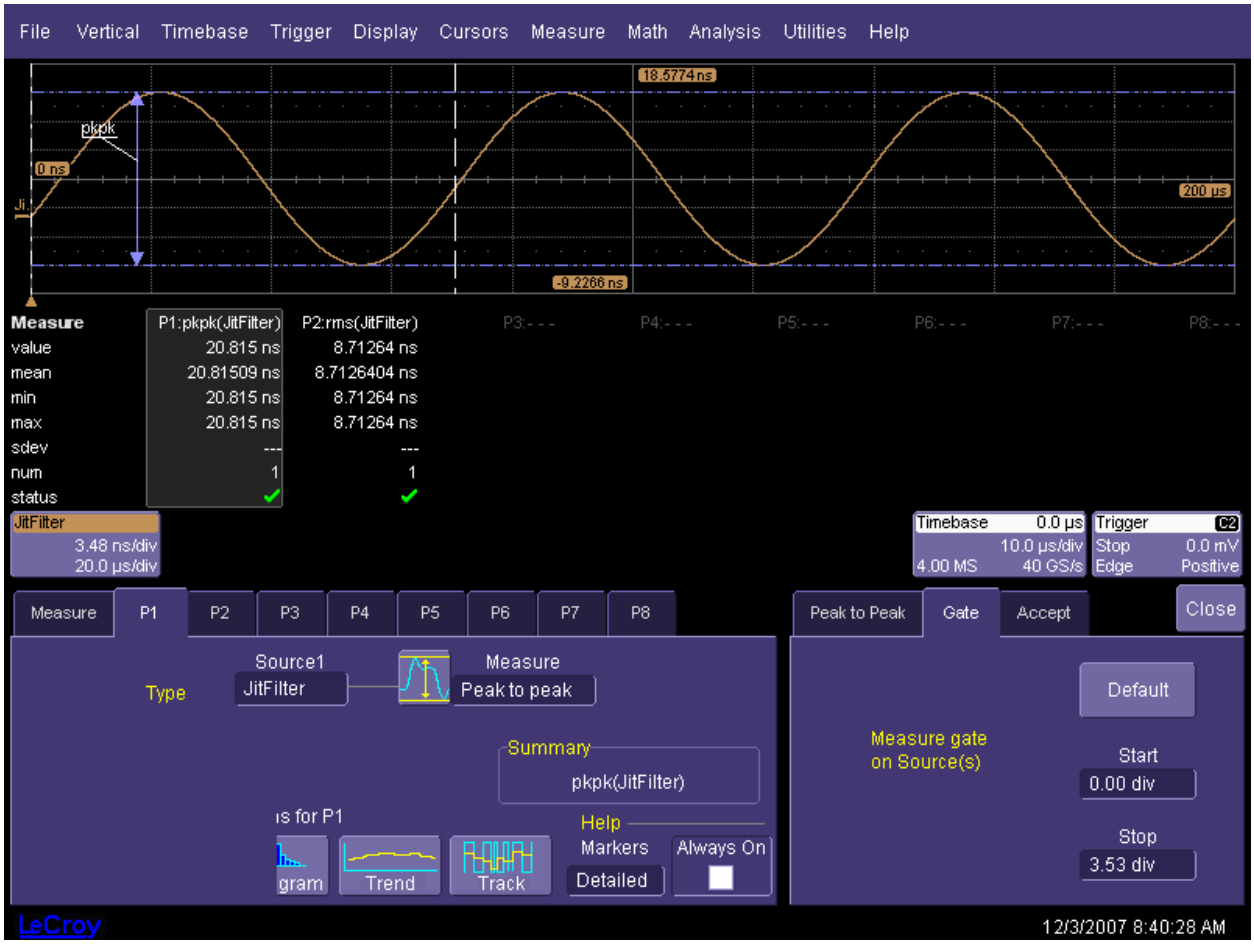
5. Click on the **PLL Settings** tab and uncheck **PLL On**.
6. Return to the **Clock Analysis** tab and click **TIE Jitter**. Two new tabs will appear: **Jitter** and **Adv. Control**.
7. Click on the **Jitter** tab. Click **Filtered Jitter** and **Advanced** and uncheck **Enable Jitter Filter**.



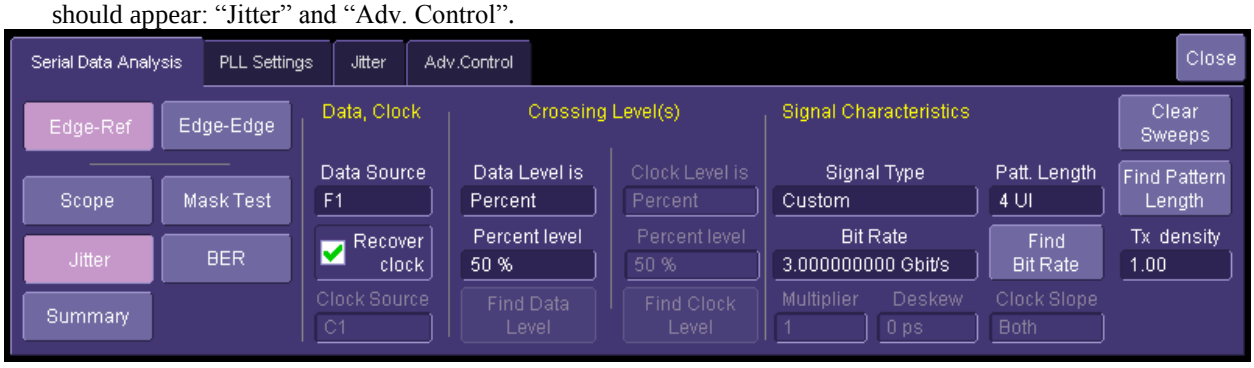
8. Press **Single** on the Front Panel to acquire one acquisition. The modulation waveform should be visible.



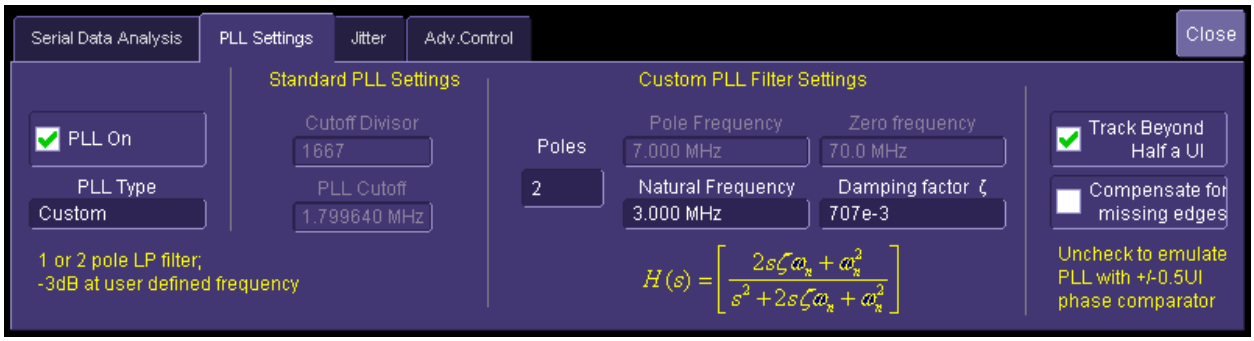
9. Click on **P1:pkpk(JitFilter)** to open the P1 measure dialog. Adjust the measure gates on P1 to surround one cycle of the modulation.



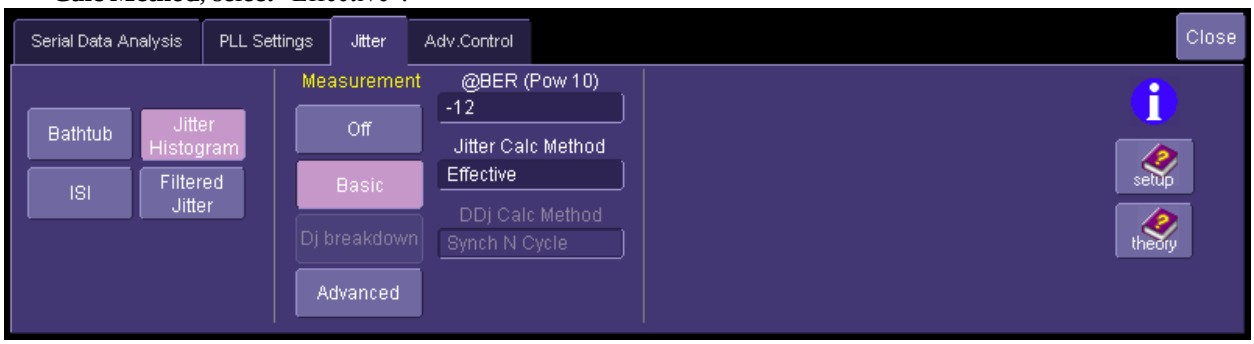
10. Verify the peak to peak level of modulation in P1: pkpk(JitFilter) is 20.8 ns +/- 10 % for Gen1 and Gen2 or 1.0 ns p-p +/- 10% for Gen3. If it is not, then adjust the output of the generator and reacquire until it does. Try to get the value as close to as possible. When satisfied, record the measured value as **DJSSC**.
11. Without making any other modifications to the generator output, turn off the sinusoidal phase modulation.
12. On the oscilloscope menu click on **Analysis** -> **Serial Data**. When prompted to recall setup click "Continue with current scope settings".
13. On the **Serial Data Analysis** tab set **Bit Rate** to "1.5Gbit/s" for Gen1i, "3.0 Gbit/s" for Gen2i or "6.0 Gbit/s" for Gen3i and **Patt Length** to "4". Click **Jitter** to turn on Jitter Measurements. Two new tabs should appear: "Jitter" and "Adv. Control".



14. Click on the **PLL Settings** tab and check "PLL On". Set **PLL Type** to "Custom." Set **Poles** to "2". Set **Natural Frequency** to "4.2 MHz" for Gen1 and Gen2 or "7.6MHz" for Gen3 and set **Damping Factor ξ** to "0.707" for Gen1 and Gen2 or "0.780" for Gen3. Uncheck **Compensate for missing edges**.



15. Click on the **Jitter** tab. Click the "Jitter Histogram" button. Under "Measurement" click "Basic." For **Jitter Calc Method**, select "Effective".



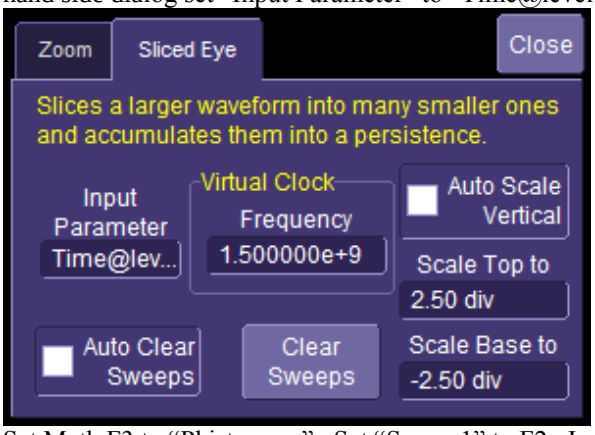
16. Set the Trigger Mode to "Normal" and wait for the scope to acquire multiple acquisitions (allow at least 1 million hits to accumulate in the histogram) then Stop the acquisition. Record the Dj value in P3 as **DJSSCOFF**.
17. Turn on the sinusoidal phase modulation.
18. On the oscilloscope, press "Clear Sweeps" on the front panel. Set the Trigger Mode to "Normal" and wait for the scope to acquire multiple acquisitions (allow at least 1 million hits to accumulate in the histogram) then Stop the acquisition. Record the Dj value in P3 as **DJSSCON**.
19. Calculate and record the level of measured DJ by subtracting the DJ with modulation off from DJ with modulation on, **DJMSSC = DJSSCON - DJSSCOFF**. Calculate the jitter attenuation by 20Log(**DJMSSC / DJSSC**). This value must fall within the range of -72dB +/- 3dB for Gen1 and Gen2 or -38.2 +/-3dB for Gen3. If it does not, then click on the **PLL Settings** tab and adjust the **PLL Natural Frequency** and reacquire multiple acquisitions until it does.
20. Adjust the pattern generator for a D24.3 pattern (00110011) and modulation to produce a 50 MHz +/-1%, 0.3 UI p-p +/- 10% (200ps for Gen1i, 100ps for Gen2i or 50ps for Gen3i) sinusoidal phase modulation, also known as periodic jitter, PJ.
21. On the oscilloscope, in **Timebase** set **Time/division** to "10 ns/div".
22. Click **Analysis -> Clock Analysis…** Set **Clock Source** to "F1", **Signal Type** to "Custom", **Clock Frequency** to "375MHz" for Gen1i, "750 MHz" for Gen2i or "1.5GHz" for Gen3i.
23. Click on the **PLL Settings** tab and uncheck **PLL On**.
24. Return to the **Clock Analysis** tab and click **TIE Jitter**.
25. Click on **P1:pkpk(JitFilter)** to open the P1 measure dialog. Adjust the measure gates on P1 to surround one cycle of the modulation. Verify the level of modulation meets the requirements and record the peak to peak level, **DJM**.
26. Without making any other modifications to the generator output, turn off the sinusoidal modulation.
27. Click on **Analysis -> Serial Data**. When prompted to recall setup click "Continue with current scope settings".
28. Click on the **PLL Settings** tab and check **PLL On**.

29. On the **Serial Data Analysis** tab set **Bit Rate** to "1.5Gbit/s" for Gen1i, "3.0 Gbit/s" for Gen2i or "6.0Gbit/s" for Gen3i and **Patt Length** to "4". Click **Jitter** to turn on Jitter Measurements. Two new tabs should appear: "Jitter" and "Adv. Control".
30. Click on the **Jitter** tab. Click the "Jitter Histogram" button. Under "Measurement" click "Basic." For **Jitter Calc Method**, select "Effective".
31. In **Timebase** set **Time/division** to "10 us/div".
32. Set the Trigger Mode to "Normal" and wait for the scope to acquire multiple acquisitions (allow at least 1 million hits to accumulate in the histogram) then Stop the acquisition. Record the Dj value in P3 as **DJMOFF**.
33. Turn on the sinusoidal phase modulation on the signal generator.
34. Press "Clear Sweeps" on the scope Front Panel. Set the Trigger Mode to "Normal" and wait for the scope to acquire multiple acquisitions (allow at least 1 million hits to accumulate in the histogram) then Stop the acquisition. Record the Dj value in P3 as **DJMON**.
35. Calculate the difference in reported DJ for these two cases, **DJMM = DJMON - DJMOFF** Calculate the -3dB value: **DJ3DB** = **DJMM** * 0.707.
36. Adjust the frequency of the PJ source to 2.1MHz for Gen1i or Gen2i and 4.2MHz for Gen3i. Measure the reported DJ difference between PJ on versus PJ off **DJ = DJON - DJOFF** and compare to the (DJ -3dB) value, **DJ3DB**. Shift the frequency of the PJ source until the reported DJ difference between PJ on versus PJ off is equal to (DJ -3dB). The PJ frequency is the -3dB BW of the JTF; record this value **F3DB**.
37. On the oscilloscope, click on the **PLL Settings** tab and adjust the **PLL damping factor** to bring the PJ –3dB frequency to 2.1MHz +/- 1MHz for Gen1i and Gen2i or 4.2MHz +/-1 MHz for Gen3i. Repeat steps 17 through 37 until both the jitter attenuation and 3dB frequency are in the acceptable ranges.
38. Check the peaking of the JTF. Adjust the pattern generator for a D24.3 pattern and modulation to produce sinusoidal phase modulation (PJ) at the –3dB BW frequency found above, and 0.3 UI p-p +/- 10% (200ps for Gen1i, 100ps for Gen2i or 50ps for Gen3i). Increase the frequency of the modulation to find the maximum reported DJ; it is not necessary to increase beyond 20MHz. Measure the reported DJ difference between PJ on versus PJ off, **DJPK = DJPKON - DJPKOFF**. Record this DJ difference (**DJPK**) and frequency, **F3PK**.
39. Calculate the JTF Peaking value: 20Log (**DJPK** / **DJMM**). Record this value.

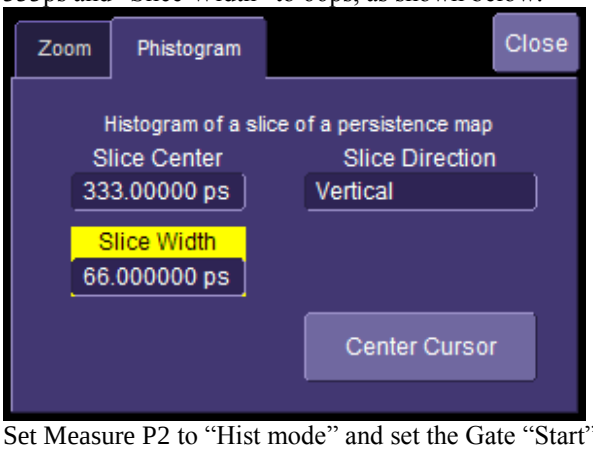
Appendix F – OOB-01 Amplitude Calibration Procedure

Test OOB-01 requires the use of a signal generator to output the OOB waveform. The generator output must be set to output 210mV for the high threshold and 40mV (1.5Gb/s products) or 60mV (3.0Gb/s products). In order to calibrate signal generator for these levels attach the generator outputs to C2 and C3 on the oscilloscope and follow these instructions:

1. Recall Default Setup.
2. Turn off Channel 1 and Turn on Channel 3.
3. In the Timebase dialog "Time/Division" to 500ns. Set "Delay" to -1.0µs. Set the sampling rate to the max sample rate.
4. In the Trigger dialog set Trigger Mode to Edge. Set "Trigger Source" to C2 and "Holdoff by:" to 5.0µs.
5. Set the "Probe Atten" for Channels 2 and 3 to the appropriate value for the attenuators attached (1.995 for 6dB), then set Channels 2 and 3 to 50mV/div.
6. Set Math F1 "Operator1" to "Difference". Set "Source1" and "Source2" to "C2" and "C3", respectively. Turn on F1 and turn Channels 2 and 3 off.
7. Set Measure P1 to "Time@Level". Set "Source1" to F1. In the right hand side dialog set "Slope" to both and "Hysteresis" to 1.0.
8. Set Math F2 "Operator1" to "Slice2Persist". Set "Source1" to P1 and set "Source2" to F1. In the right hand side dialog set "Input Parameter" to "Time@level" and "Frequency" to 1.5GHz as shown below:



9. Set Math F3 to "Phistogram". Set "Source1" to F2. In the right hand side dialog set "Slice Center" to 333ps and "Slice Width" to 66ps, as shown below:



10. Set Measure P2 to "Hist mode" and set the Gate "Start" at 5 divisions. Set "Source1" to F3. This measures the mode of the upper level of the signal between 45% and 55% of a UI.
11. Set Measure P3 to "Hist mode" and the set the Gate "Stop" at 5 divisions. Set "Source1" to F3. This measures the mode of the lower level of the signal between 45% and 55% of a UI.

12. Set Measure P4 type to “math on parameters” and set the math operator to P Difference. Set “Source1” to P2 and “Source2” to P3. This is the differential voltage.
13. Turn on Measures statistics.
14. Set the oscilloscope to Normal trigger.
15. Adjust the generator output voltage to 210mV. Press clear sweeps and observe the value of P4 mean. If it does not equal 210mV readjust the voltage and press clear sweeps again.
16. Repeat step 14 until the correct generator settings are determined. Record the settings and repeat for 40mV and 60mV.

